CARC NTPclock Kit Build Instructions

PCB Version 20261019

Software Version 4.x



Many thanks to Richard (VK2ARH), who put a monumental effort into creating the original version of these instructions.

These NTPclock Kit Build Instructions (specific to the NTPclock v4 for CARC project kit), documentation, and latest firmware for this kit can be downloaded from:

https://github.com/mjculross/NTPclock-for-MWRS-CARC-REAST

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Introduction

The NTPclock project uses the Network Time Protocol (NTP) to display accurate date and time, which is periodically updated via an internet connection to ensure that the clock maintains an accurate time display. When (re)started, or when time changes resulting from the move to and from Daylight Saving Time, the NTPclock automatically sets and adjusts the time for your chosen time zone(s), while also constantly displaying UTC.

The **Network Time Protocol (NTP)** is a networking protocol designed for clock synchronization between computer systems over packet-switched, variable-latency data networks. Developed by David L. Mills at the University of Delaware in 1981, NTP is one of the oldest internet protocols still in use.

This NTPclock project is an extension of the original NTP Clock project developed by Bruce Hall (W6BH), and a wealth of information relating to his initial project can be found on his website: http://w8bh.net/. Following the release of Bruce's initial project, a significant number of hardware and software enhancements have been undertaken, and at the time of writing this document, the clock now delivers the following features:

- Accurate time source which automatically changes with Daylight Saving Time
- Multiple time zones and UTC time displays
- Solar weather data
- Local environmental data (temperature, humidity, pressure) with configurable metric/imperial display
- Local weather data using current location data
- WiFi signal strength and status indicator
- Brightness adjustment and night mode display
- Configurable color schemes, 12/24hr, AM/PM, date format
- AP Mode available for web-based dialogue and configuration
- Live GPS data (Latitude/Longitude, Elevation etc.) – also used to define weather location and pressure calculations
- GPS clock time sequencing – substituting for NTP as the time source when WiFi is unavailable
- Automatic calculation of maiden head grid square location
- LiPo battery management and support
- 5VDC power supplied via USB
- Battery voltage sensing/display
- Software upgradeable
- Future display options limited by imagination and time ☺

The current software release v4 contains significant upgrades to the delivered functionality, including major enhancements to the web based configuration tools using the NTPclock wireless Access Point (AP). The software continues to be developed to improve and deliver additional functionality.

A full list of features and capabilities available in the software can be found in the software operations manual, together with instructions to assist in uploading the latest firmware to the processor. While earlier versions of the NTPclock used different

processors, the latest software enhancements now work best when running on the ESP32-S3-mini module processor board, with its increased processing power, memory and I/O ports.

Kit Assembly

Equipment Needed:

The following equipment is recommended to build the kit:

- Soldering iron / station (preferably ESD protected with a fine tip, together with a method of venting fumes away from the soldering area).
- Solder: 0.8mm (0.031") or smaller diameter 63% Sn / 37% Pb with 'no clean' flux is recommended.
- Solder flux can be an advantage in soldering some components, but is not essential if a good quality solder with embedded flux is used.
- Wire cutters (fine precision type)
- Multimeter
- Phillips head screwdriver
- Modeling knife
- Solder sucker or solder wick if needed.
- A magnifying glass or higher power reading glasses can assist with checking the quality of your soldering.
- A plastic container to house the component parts so as not to lose them during construction.
- If you have one available, a current limited lab power supply can assist with power up testing. Note: this is a desirable feature, not a requirement.

Take the time to prepare your work area and layout your tools, keeping your area free of clutter. An organized work area helps you to focus on the build and not be distracted by searching for tools and components.



Richard's (VK2ARH) example of a cleared and ordered work bench ahead of the kit build. Individual equipment will vary. (He readily admits that his workbench isn't always that tidy ☺)

The hardware design and software for this project are all open source, so that individuals and radio/electronics clubs can easily assemble their own ‘kits’ to build this project either individually or as a group project. The kits are not sold commercially. If you develop additional functionality, please let Richard (VK2ARH) know and publish it so that others can similarly leverage from the activity. His email contact details can be found at QRZ.com under VK2ARH.

Bill of Materials

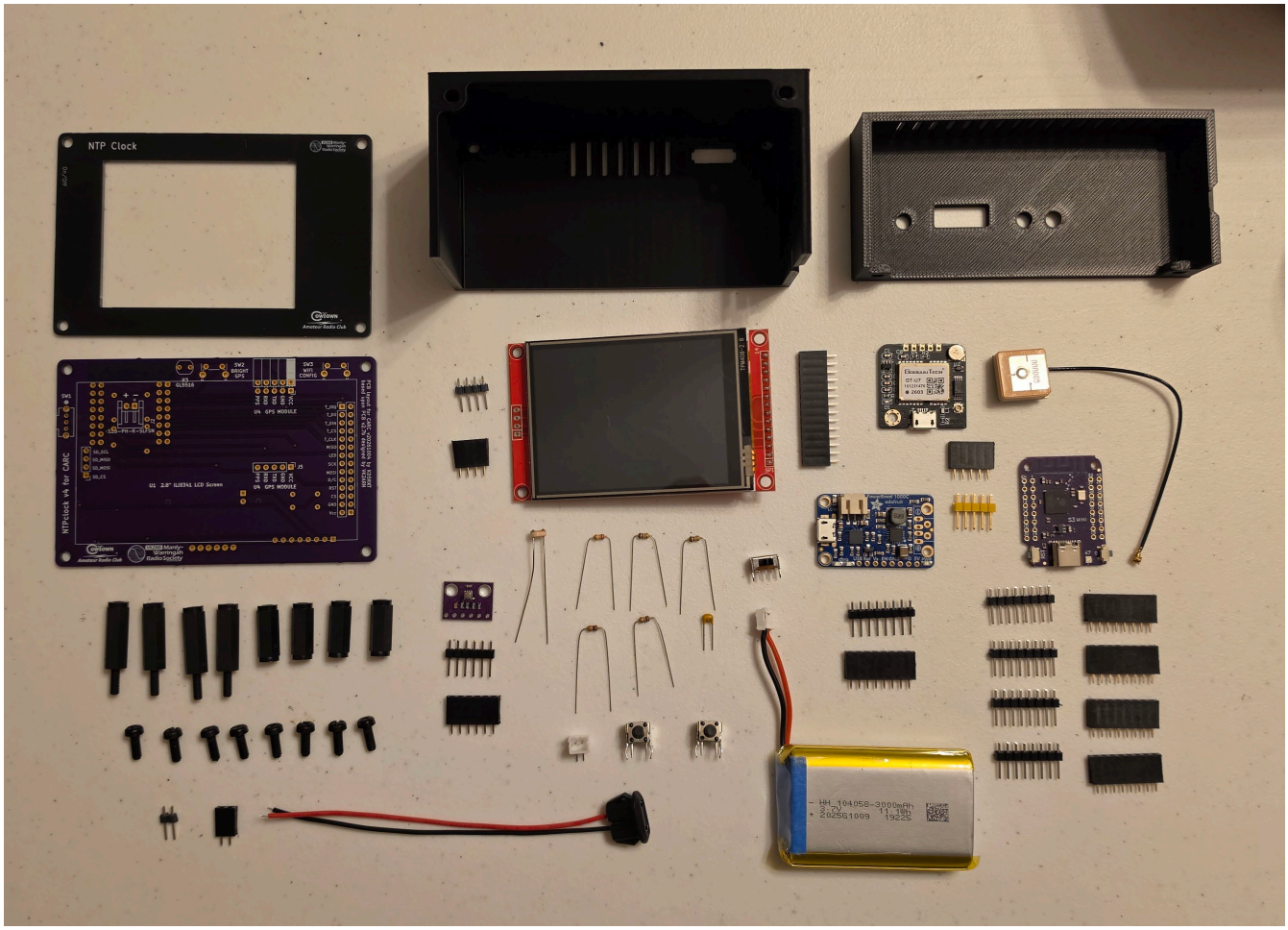
✓	Qty	RefDes	Description
	1	C1	0.1uF capacitor (104)
	1	J2	PH 2.0mm battery socket
	1	J3	USB-C panel-mount pigtail
	2	R1, R5	100K resistor (BRN-BLK-YEL)
	1	R2	27K resistor (RED-VLT-ORG)
	1	R3	GL5516 LDR
	1	R4	10K resistor (BRN-BLK-ORG)
	1	R6	220K resistor (RED-RED-YEL)
	1	SW1	SPDT on/off power switch
	2	SW2, SW3	Momentary pushbuttons
	1	U1	2.8" ILI9341 LCD screen
	1	U2	Adafruit PowerBoost 1000C battery management module
	1	U3	WEMOS ESP32-S3-mini module
	1	U4	GT-U7 GPS module
	1	U5	BME280 sensor
	4		M3 17mm F-F Nylon Insulated Standoffs
	4		M3 20mm M-F Nylon Insulated Standoffs
	8		M3 8mm Nylon Screws
	As Req		0.1" male pin header strips
	As Req		0.1" female socket header strips
	Optional		0.1" female 5-pin right-angle socket strip
	Optional		2500mAh to 3000mAh LiPo battery (optional)

NOTES:

- LiPo battery support enables free standing operation and is particularly useful when operating remotely (e.g. in the field). A limited quantity of pre-wired LiPo batteries (3000mAh capacity) will be made available for purchase (cost: $6.00) on the day of the CARC kit build-a-thon activity. Support for battery operation requires the installation of the optional Adafruit PowerBoost 1000C battery management module.
- Several GPS header socket options are provided on the PCB to support a removeable and/or permanently installed GPS module. A right-angle female socket header strip for temporary installation of the GPS is not included in the kit, but will be made available (free) on the day of the CARC build-a-thon activity.
- The number of male pin header strips and female socket header strips required for installation of all items will be determined by the options chosen. The standard 0.1” male pin header strips and female socket header strips provided can be cut down into enough appropriately sized smaller headers to support all modules chosen for installation.

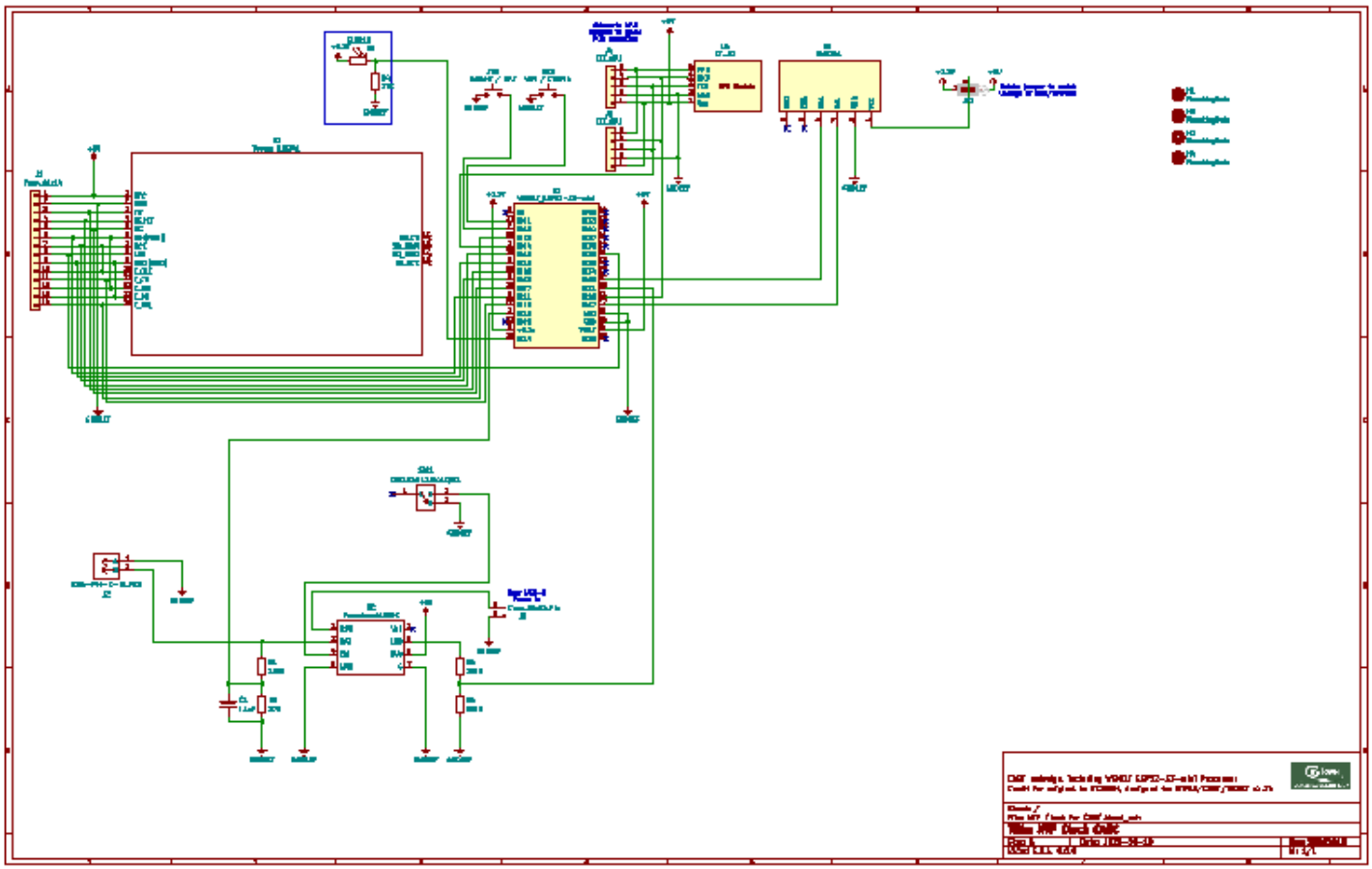
Kit Composition

Start by unpacking your kit, identifying each component, and checking the components against the BOM.



Example (not necessarily complete for this build) of the NTPclock kit components. Note: The 0.1” male pin header strips & socket header strips supplied will need to be cut down into specific sizes as required.

CARC NTPclock v4 (20261019) Schematic



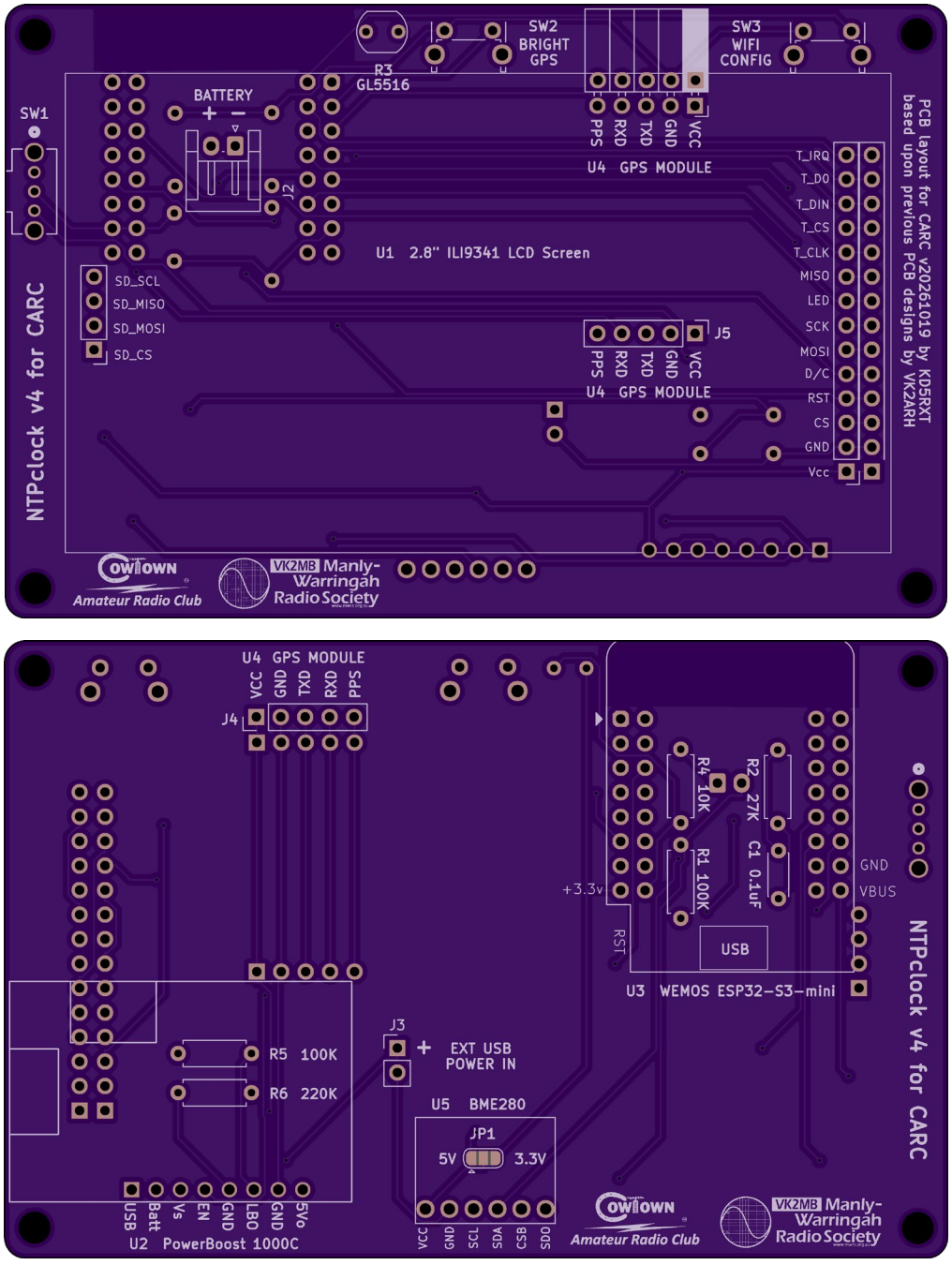
Building the CARC NTPclock kit

Assembly of the NTPclock takes approximately one hour, and could be considered an easy project for beginners with some experience soldering. Assembly for a newcomer just learning to solder (under supervision) is likely to take just a little longer. These build notes assume that you are familiar with soldering components onto a PCB.

The ‘kit’ provides all the components necessary to build the NTPclock which uses a 5VDC USB-C external power supply, and includes push button switches to support direct entry of some clock configuration options, and the circuitry to support automatic brightness adjustment and battery voltage monitoring. The components to support LiPo battery operation and GPS operation are also included, but are not absolutely required to be installed in order to operate the base USB-C powered clock. These optional components can be added by following along with these instructions, but their installation is not necessarily sequence dependent.

Main PCB – Front and Rear View

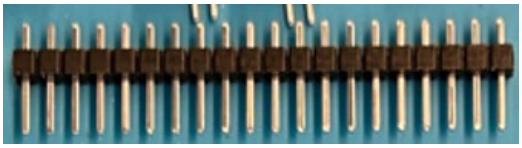
NTPclock v4 for CARC (v20261019) PCB:



Construction Guide

General Guidelines

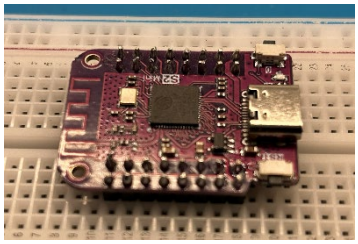
It is strongly recommended that you **solder ONLY the resistors, capacitor, and battery connector directly onto the main PCB.** All other components (ESP32-S3-mini processor module, Adafruit PowerBoost 1000C, BME280, ILI9341 TFT LCD display, etc.) will be installed using female socket headers soldered into the main PCB and corresponding male pin header strips soldered onto these components. This gives you the flexibility to easily replace these components should it be required. Installing the GPS (if chosen for inclusion on your individual kit build) allows for several options: installation of a female socket header to allow optional removal of the GPS module, permanent installation of the GPS module on the front of the PCB, or permanent installation of the GPS module on the rear of the PCB. Actual installation of the GPS will be covered later in this guide.



The kit is supplied with male pin header strips which can be snapped or cut into smaller sections as required by the project. When soldering male pin header strips, remember that the pins are held in position by a plastic housing which can easily melt if too much heat is applied. A good technique is to solder quickly and spread the heat load, soldering every second pin and then returning to pick up the unsoldered pins. It is also important to ensure that the header pins are installed perpendicular to the module and/or PCB. **Take care not to damage or make contact with any existing components on the module with your soldering iron when soldering the male pin header strips to the modules.**

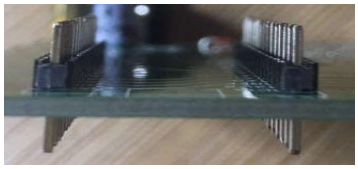
The quickest method to do this is to insert the appropriately sized strip of male header pins into the module (in the correct orientation) and secure the strip by soldering a single pin. You then visually align the strip while melting the solder on the pin and reposition the strip holding it in position while the solder cools. Repeat as necessary until the strip is aligned correctly, then proceed to solder the remaining pins to attach the strip to the module.

If you don't wish to use this method there are a couple of techniques that can assist in ensuring the perpendicular installation of the header pins.



The first involves the use of a prototype breadboard - place the module and all the required pins into place, ensuring vertical alignment of the pins and then soldering the module to the pins.

If using this technique, be careful not to distort the male pin header strips inserting them into the breadboard as some of the holes in the breadboard can be difficult to insert, which will misalign the male pin header strips in the vertical plane. If this occurs, level up the pins before soldering.

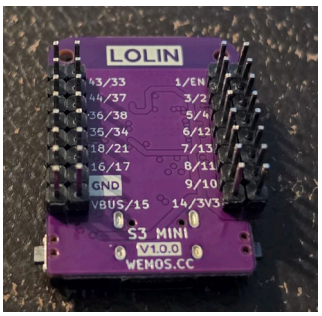


Perhaps the easiest way is to place the male pin header strip into the main PCB (long lead into the board) to ensure correct alignment. Place the PCB down on a support to lift it up so that the pins are not pushed upward when placed down for soldering, set the module onto the short side of the male pin header strips, component side up, and carefully solder the male pin header strips to the module. Remove the module with the male pin header strips connected from the PCB for later installation onto the board, after the female socket header strips have been installed onto the PCB.

The female socket header strips used in the project are generally supplied as a strip of headers which need to be cut to the required length for each application. The easiest way to do this is to carefully measure the number of pins required, measure twice – cut once. Remove one sacrificial pin before making the cut to make it much easier to cut the plastic. Using a pair of wire

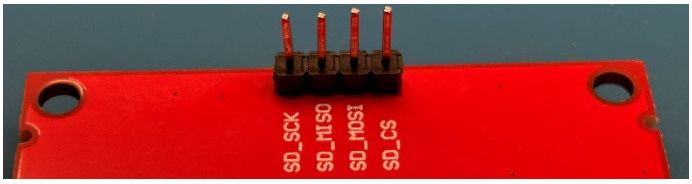
cutters, carefully cut the strip (where the sacrificial pin was removed) to the required length, leaving the required number of pins in the strip as needed. Once cut to length, the end of the strip can be sanded or carefully trimmed with a modelling knife to achieve a smoother edge (although this is only for cosmetic appearance).

Preparing the ESP32-S3-mini Processor Module

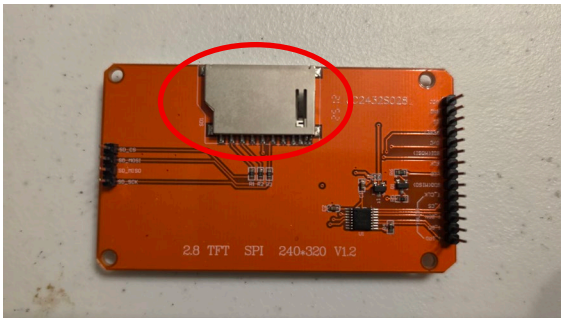


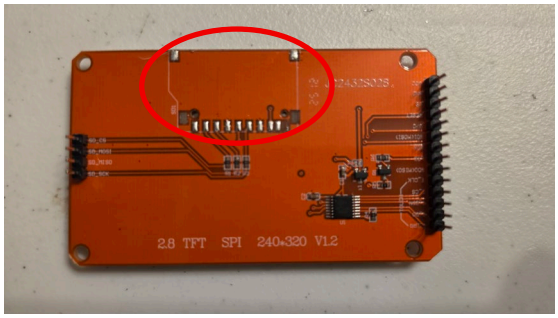
- Solder the four 8-pin male header strips to the ESP32-S3-mini processor module using the techniques described above. The finished board should look like that shown on the right. The longer unsoldered legs of the male pin header strips are pointing upwards away from the bottom of the module.

Preparing the Display Module



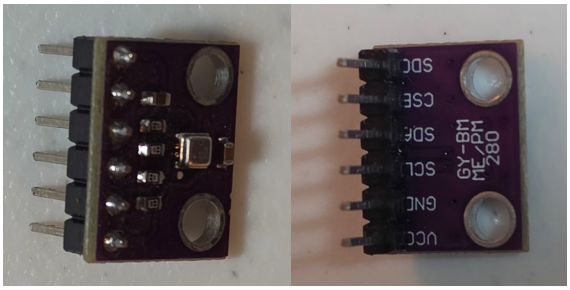
- Prepare and solder a 4-pin male header strip onto the display using the techniques described above. The finished board should look like that above. The longer unsoldered legs of the male pin header strips are pointing upwards away from the bottom of the module. The display is normally already fitted with the male 14-pin header strip. The male 4-pin header strip just installed is used simply to provide mechanical stability when the display is mounted on the PCB.





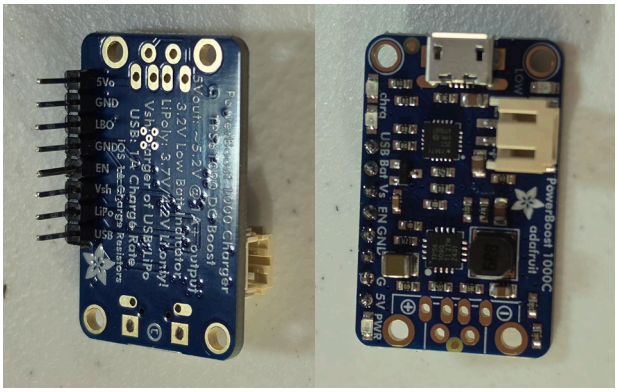
- In order to eliminate any potential interference with an optional battery (to be installed later), carefully remove the SD socket from the rear of the TFT display, as indicated by the red ovals in the “before” and “after” pictures above by unsoldering all of the connections securing it to the TFT display PCB. Do not be concerned if any PCB pads are lifted in the process of removing the SD socket, as this will not adversely affect the normal operation of the display in any way.

Preparing the BME280 Environmental Module



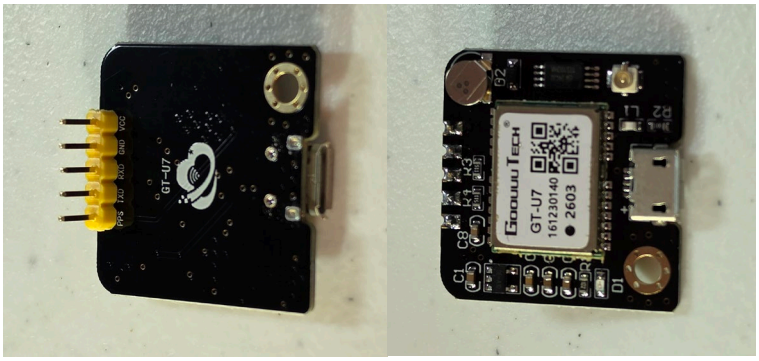
- ☐ Prepare and solder a 6-pin male header strip onto the BME280 module using the techniques described above. The finished board should look like that above. The longer unsoldered legs of the male pin header strips are pointing upwards away from the bottom of the module.

Preparing the Adafruit PowerBoost 1000C Battery Management Module



- ☐ If battery operation is desired, prepare and solder a 8-pin male header strip onto the Adafruit PowerBoost 1000C battery management module using the techniques described above. The finished board should look like that above. The longer unsoldered legs of the male pin header strips are pointing upwards away from the bottom of the module.

Preparing the GPS Module



- If GPS operation is desired, prepare and solder a 5-pin male header strip onto the GT-U7 GPS module using the techniques described above. The finished board should look like that above. The longer unsoldered legs of the male pin header strips are pointing upwards away from the bottom of the module.

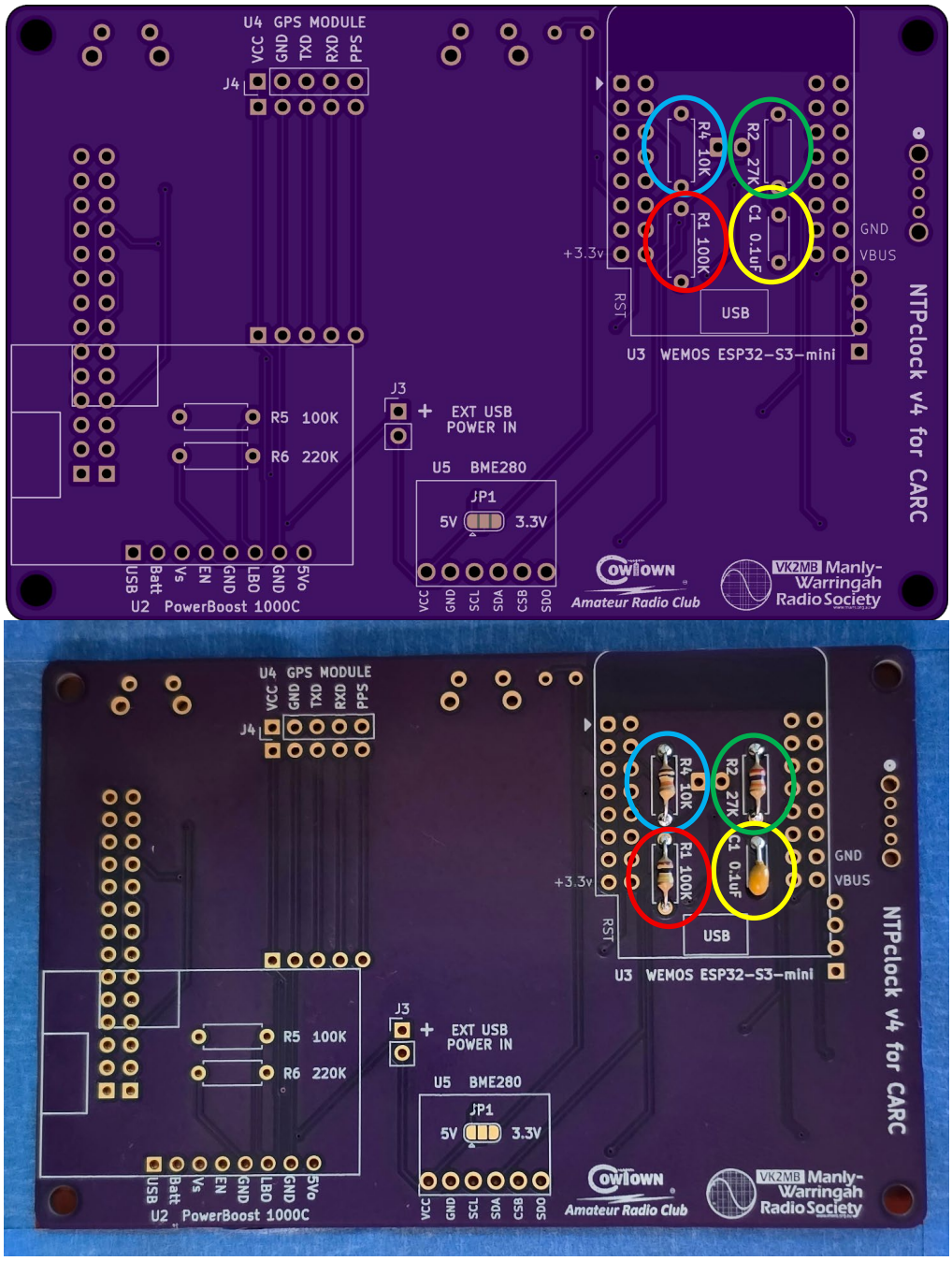
Preparing the Female Socket Header Strips

- You will need the following female socket header strips: two female 14-pin socket header strips & one female 4-pin socket header strip (to mate with the display), four female 8-pin socket header strips (to mate with the ESP32-S3-mini module), and one female 2-pin socket header strip (to mate with the rear-panel USB-C power connector, with a corresponding male 2-pin header strip on the PCB). For installing the BME280 module, you will need a female 6-pin socket header strip. Similarly, if you plan on installing the GPS module, depending upon the installation method chosen, you may need an additional female 5-pin socket header strip. And finally, you will need a female 8-pin socket header (to mate with the Adafruit PowerBoost 1000C battery management module, if chosen for installation).

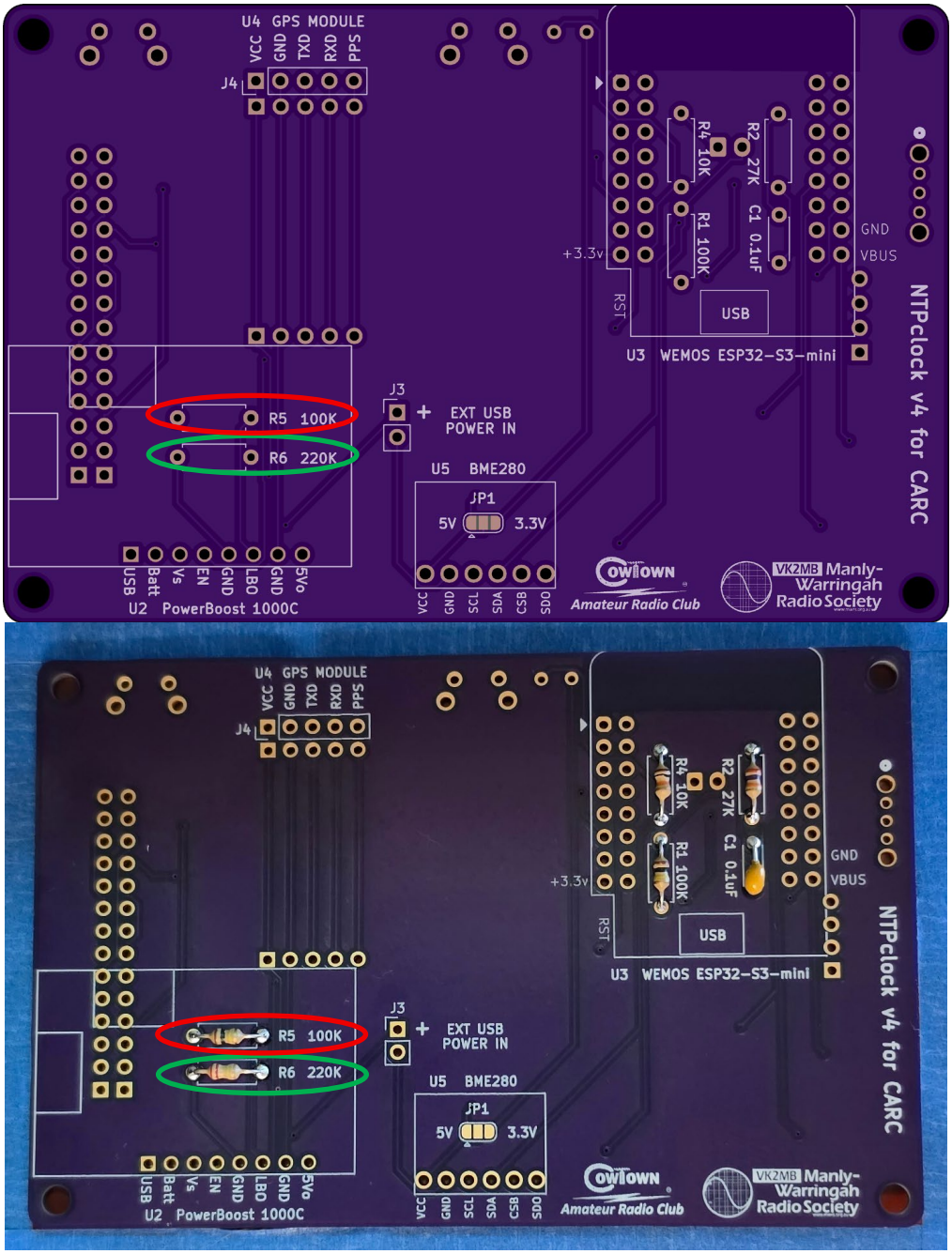
  NOTE: Do not install any female header strips into the PCB at this time. Installation of these strips will be covered later in these instructions.

Installing Thru-Hole Components

Start with the smallest and basic components:



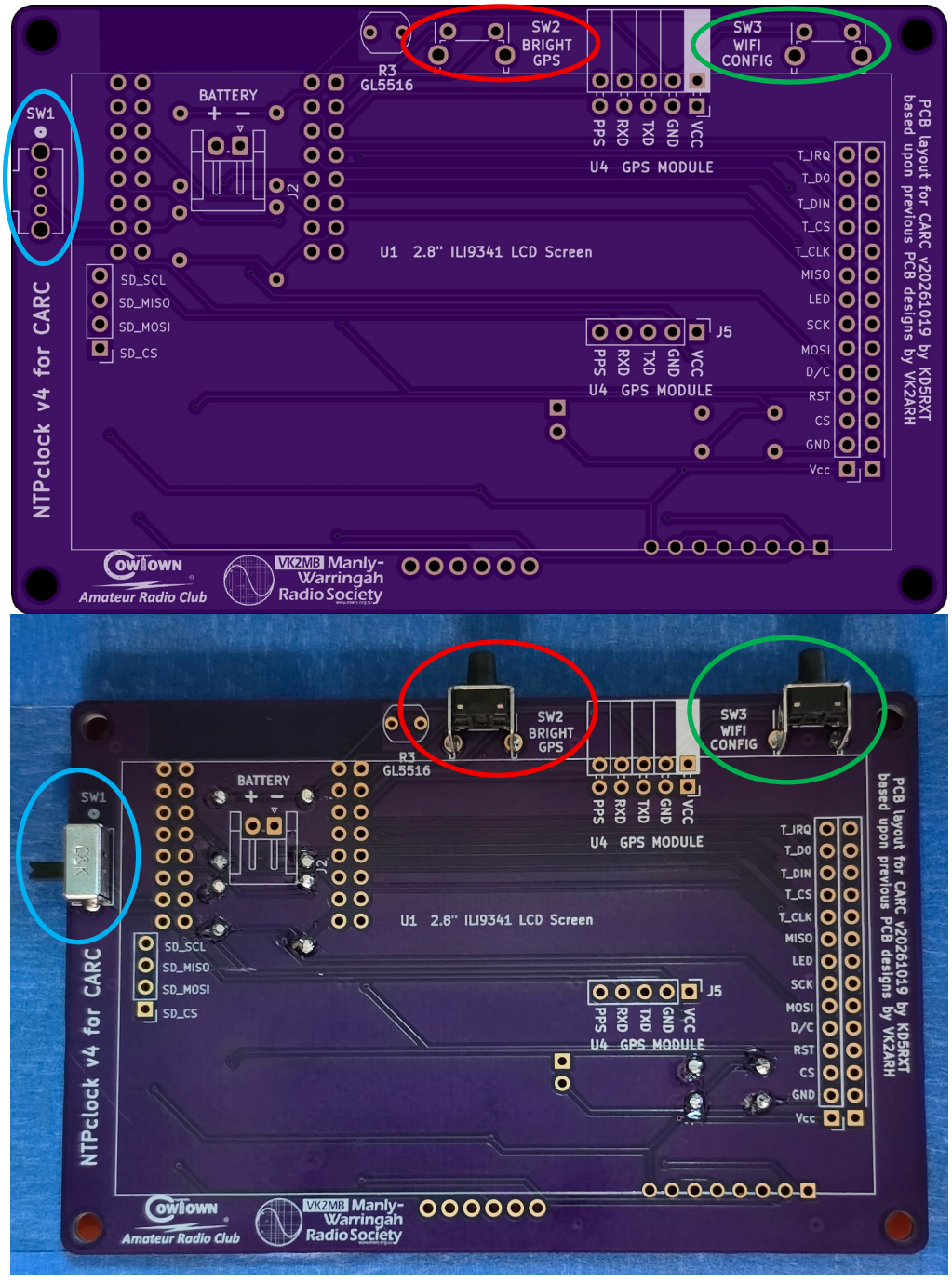
☐ [Rear of the PCB] Install R1 (100K brown-black-yellow), as indicated by the red oval above. Install R2 (27K red-violet-orange), as indicated by the green oval above. Install R4 (10K brown-black-orange), as indicated by the aqua oval above. And finally, install C1 (0.1uF marked 104), as indicated by the yellow oval above. After installation of each component, trim any excess lead length using wire cutters. Retain the trimmed leads, as they are thus available to be used later in these instructions as jumpers.



☐ [Rear of the PCB] Install R5 (100K brown-black-yellow), as indicated by the red oval above. Install R6 (220K red-red-yellow), as indicated by the green oval above.

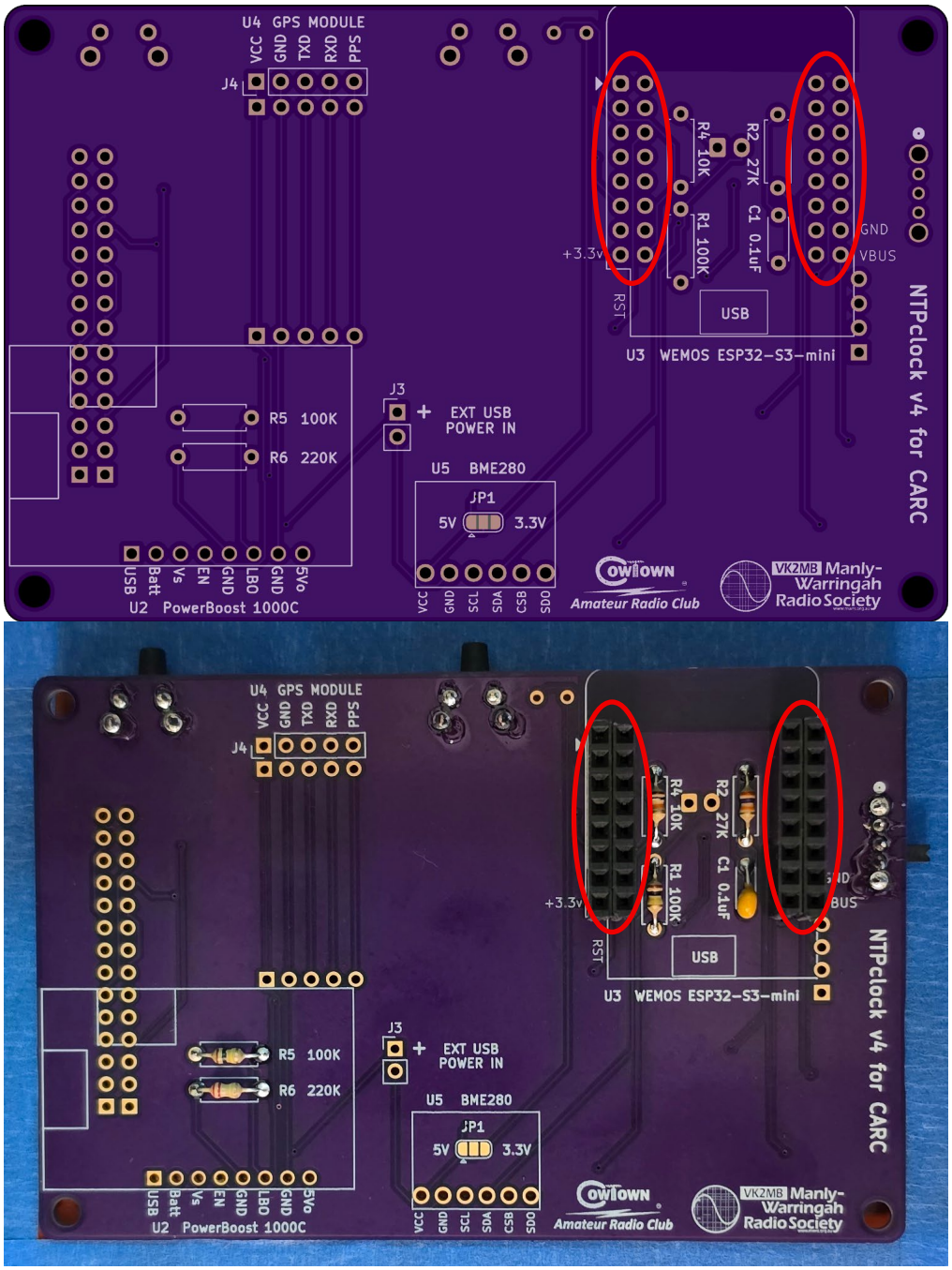
Installing Switches and Sockets

Now install some of the larger components on the front side of the PCB:



☐ [Front of the PCB] Install SW2 (BRIGHT/GPS), as indicated by the red oval above. Install SW3 (WIFI/CONFIG), as indicated by the green oval above. Install SW1 (ON/OFF – NOTE: the sliding handle of the switch should hang off of the edge of the PCB, pointing to the left in the picture), as indicated by the aqua oval above.

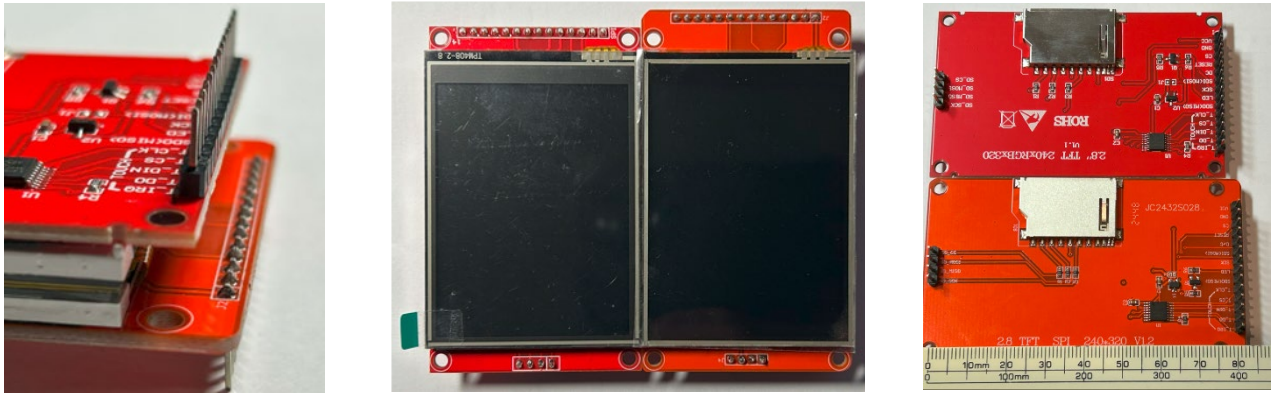
Installing Female Socket Header Strips for the ESP32-S3-mini



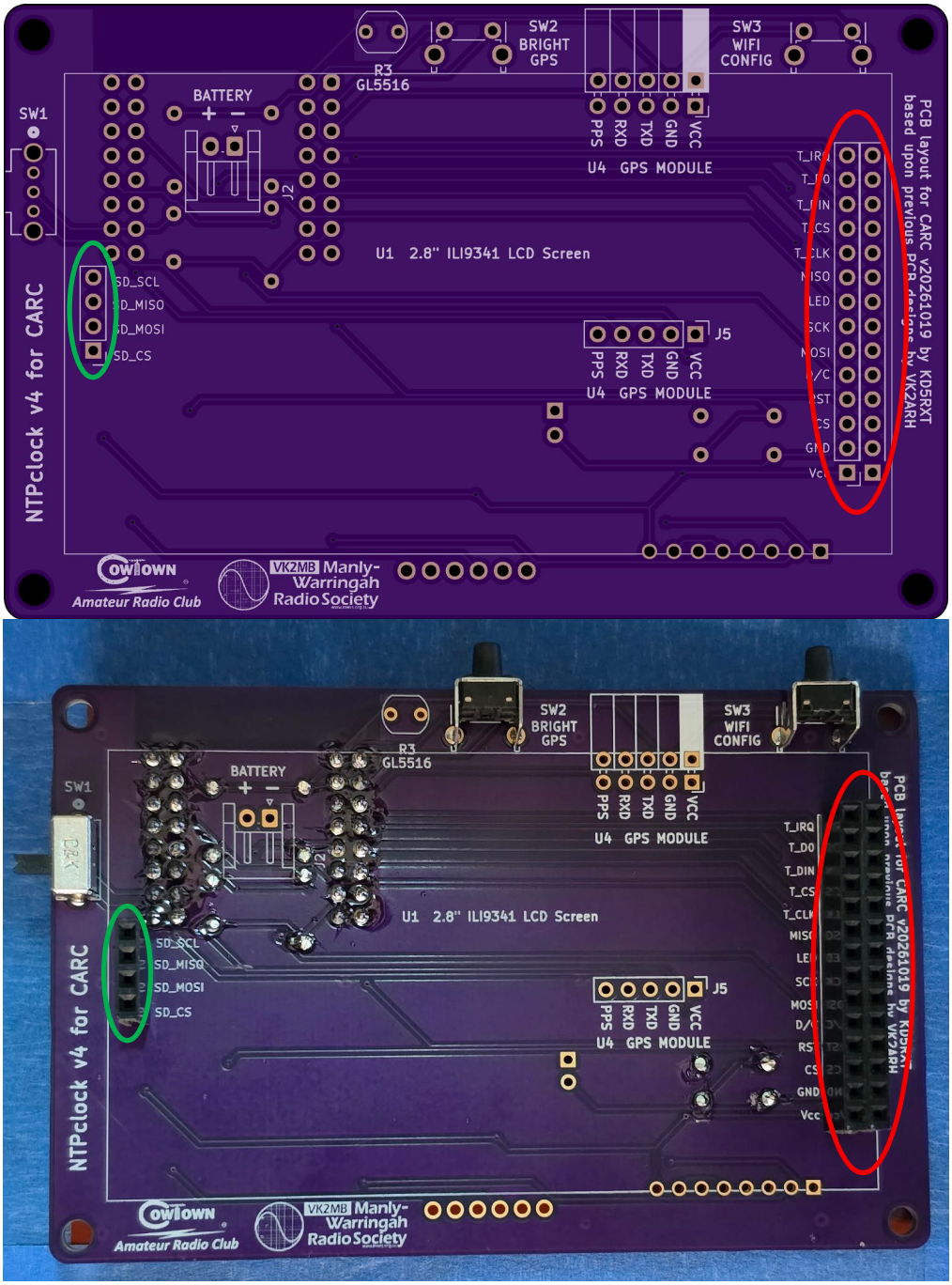
☐ [Rear of the PCB] Install the female socket header strips, as indicated by the red ovals above, for the ESP32-S3-mini module on the rear of the PCB (it's easier to solder them in place before installing the female socket header strips on the front of the PCB for the display). Rotate the board and insert the female socket header strips from the rear of the PCB and solder them from the front side of the PCB.

Installing the Socket Header Strips for the TFT Display

Different manufacturers provide 2.8” TFT screens with different male 14-pin socket connector locations. The CARC NTPclock v4 (20261019) PCB provides two sets of female 14-pin socket connector positions to enable use of either of the common pin configurations found on the 2.8” TFT displays.



Different TFT Displays have the male 14-pin header strips in different locations. The longer board meets the original factory specification.



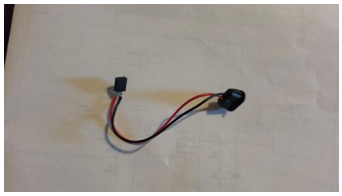
- [ ] [Front of the PCB] Install the two 14-pin female header strips, as indicated by the red oval above, into the two connectors on the front of the PCB (soldering them in place from the rear of the PCB). Install the 4-pin female header strip, as indicated by the green oval above, on the front of the PCB (soldering it in place from the rear of the PCB). **Be extra careful when soldering the 4-pin female header strip to the PCB to avoid contacting the female header strips installed earlier for the ESP32-S3-mini processor module with the hot soldering iron tip.**

You may find it easier to solder the female header pin strips into position after installing them in the TFT display's male header pins, and then inserting the display into the front of the PCB, turning the PCB over and carefully resting the PCB on the display to solder the female header pins to the rear of the PCB for the connector that matches your display. The second female 14-pin

header strip can simply be aligned with the one already in place and, just as with the first, solder the female header pins to the rear of the PCB.

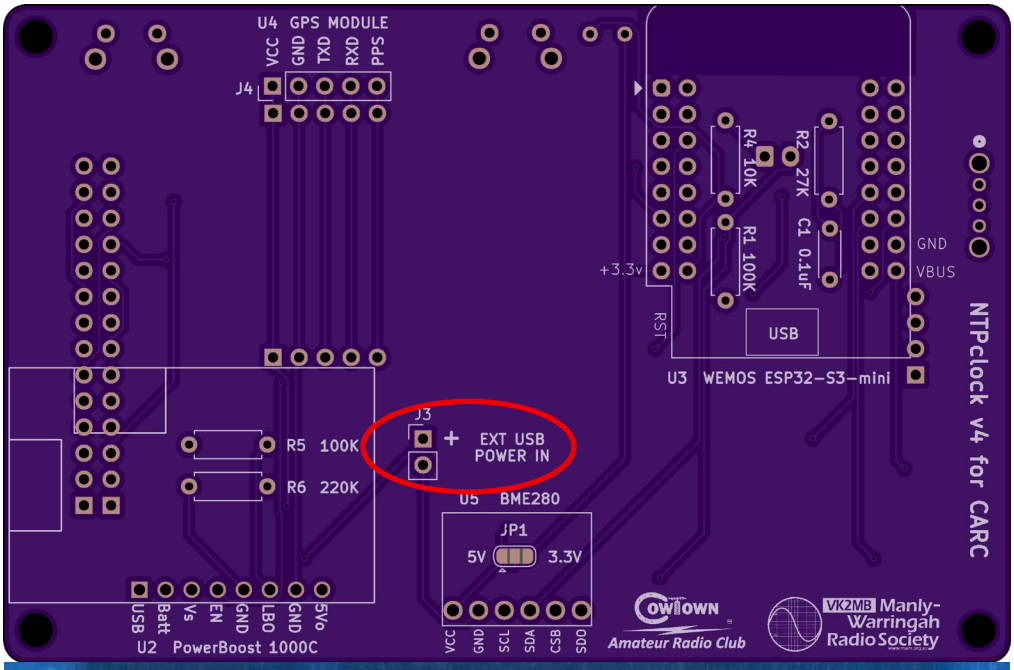
Preparing the Rear USB-C Panel Mounted power connection (or charging outlet)

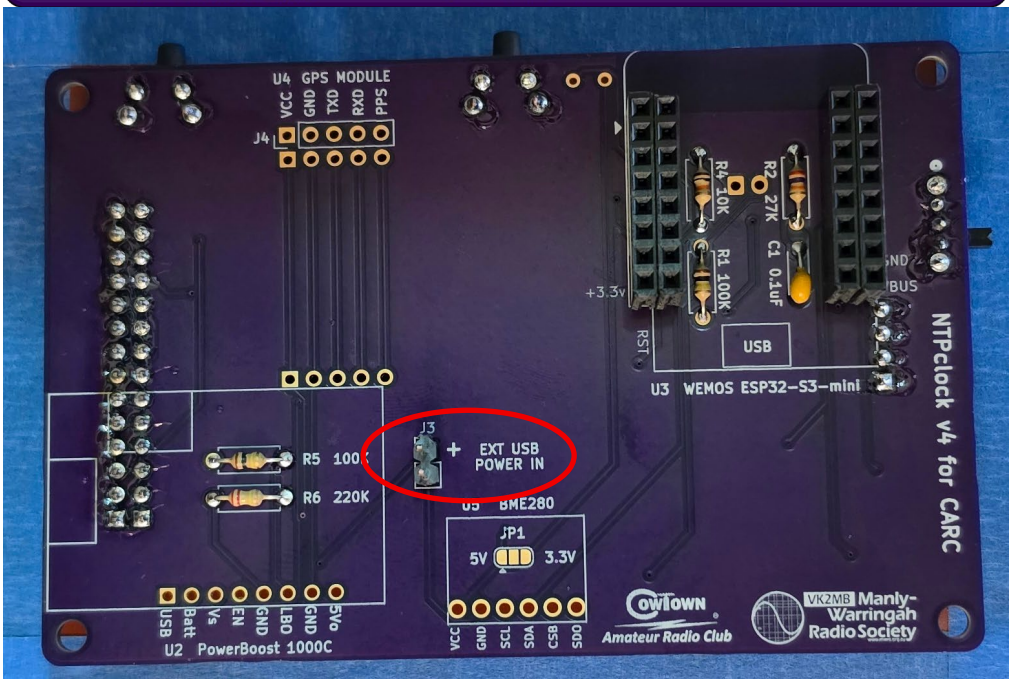
A panel mounted USB-C power socket is supplied for mounting on the rear panel of the clock's 3D-printed enclosure. This socket is handy for powering the clock from the rear, rather than from the side. This socket can also be utilized as a 5VDC power out socket, as it is wired in parallel with the microUSB connector on the battery management module (installation of the battery management module is covered later in this document).





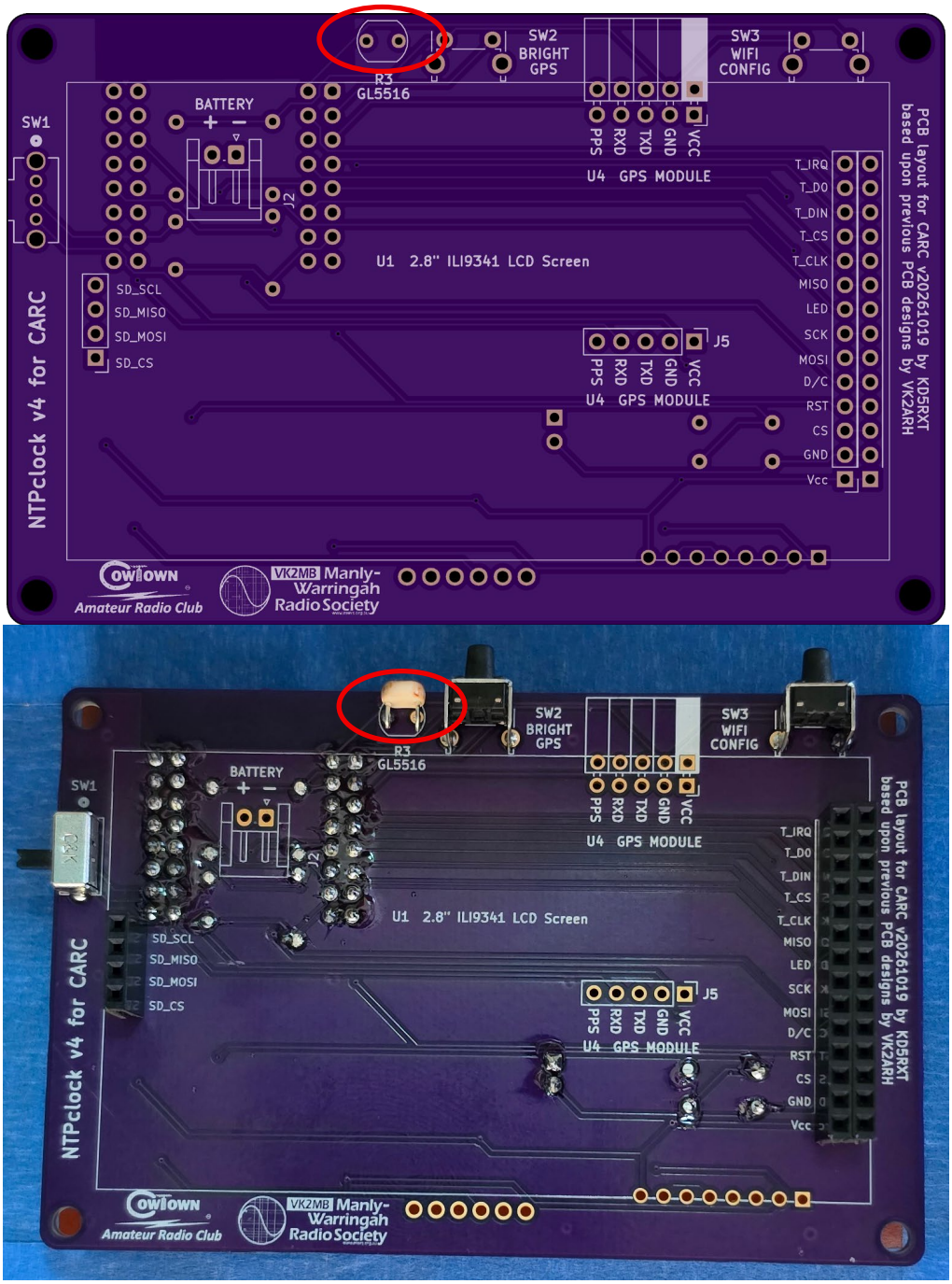
- Solder the wires on the panel mounted USB-C socket to a female 2-pin socket header, as shown in the top picture on the right (this cable will eventually be plugged into J3 on the rear of the PCB). Optionally (but recommended), hot glue (which will be made available for use on the day of the CARC build-a-thon activity) can be applied to the rear of the 2-pin socket strip to provide needed stress-relief for the wires, as shown in the bottom picture on the right. If heatshrink (not provided in the kit) is used as strain-relief for the wires, make sure that the overall length of heat-shrink is not excessive to ensure that interference with the back panel of the NTPclock 3D-printed case is not caused as a result of the length of the completed assembly.





☐ [Rear of the PCB] Install a male 2-pin header strip, as indicated by the red oval above, into the EXT USB POWER IN (J3) location on the rear of the PCB.

Installing the LDR



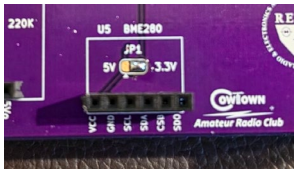
☐ [Front of the PCB] Before installing the LDR (R3 GL5516), bend the legs 90° to the board. Solder the LDR on the front of the PCB, as indicated by the red oval above, so that it is oriented facing up & aligned with the edge of the PCB as shown on the right.



If using the LDR to control the display brightness is not needed/desired, a jumper can be installed to get full brightness instead. Form a jumper using a piece of a leg trimmed from any resistor installed earlier, bent into a U-shape, then soldered in place

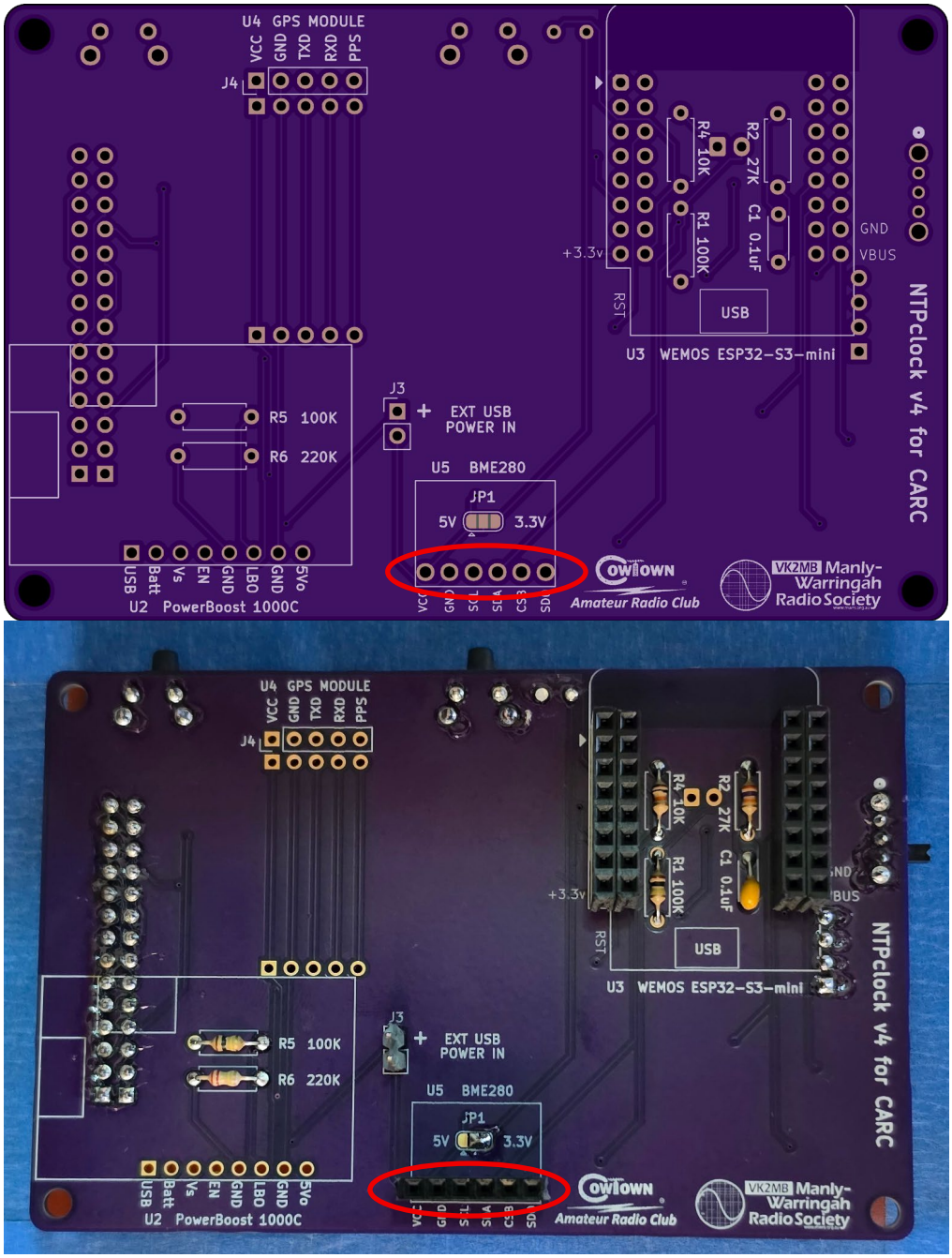
of the LDR. This will allow easy removal of the jumper at some time in the future, if desired.

Installing the Socket Header Strip for the BME280 Module



The BME280 module provides temperature, pressure and humidity readings. It is highly recommended to install the BME280 module using a female 6-pin socket header, which places it standing off of the main PCB to improve airflow around the module and reduce the impact of any PCB heating on the modules reading.

- [ ] [Rear of the PCB] Before installation, you will need to solder jumper JP1 on the PCB to select the required 3.3V supply (as shown) for the BME280 module included in the kit.



☐ [Rear of the PCB] Install the female 6-pin socket header strip for the BME280 module on the rear of the PCB, as indicated by the red oval above.

Lithium Battery Support

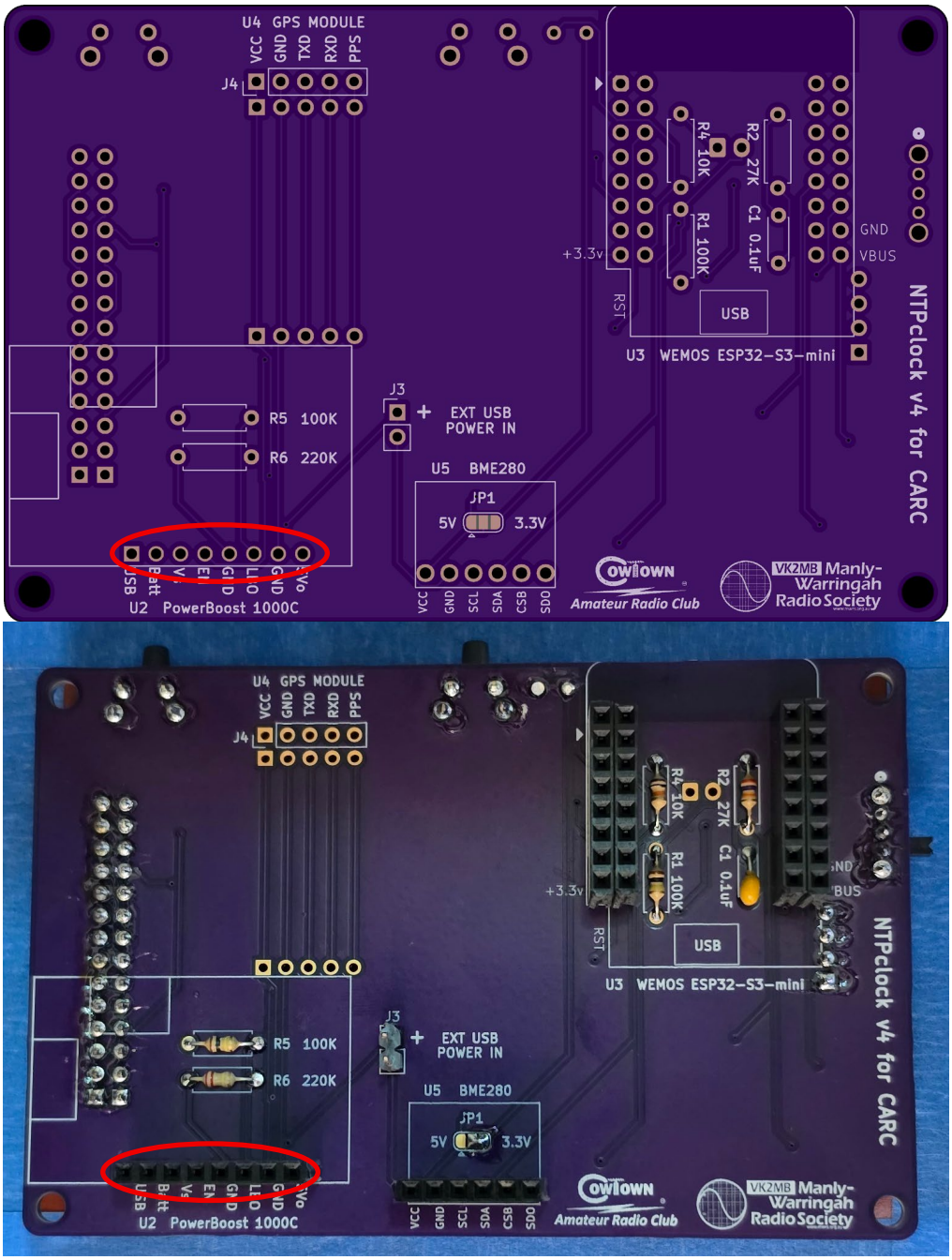
The NTPclock v4 for CARC (20261019) PCB supports operation from a 3.7v LiPo battery, using the optional Adafruit PowerBoost 1000C battery management module, a suitable LiPo battery, and an additional 2.0mm JST PH LiPo battery connector.

NOTE: This manual assumes that the builder has a competent understanding of electronics and principles associated with the use, management, and handling of LiPo batteries. The information presented here is provided as a guide, and does not abdicate the responsibility of the builder to ensure safe operation of the LiPo battery system that they install. If you are in any doubt as to your competency to undertake this task, you should seek qualified assistance for installing the LiPo battery in your project.

WARNING: The batteries and the battery charging module used in the NTPclock v4 for CARC (20261019) PCB follow the Adafruit convention for LiPo battery connection polarity. It is absolutely critical that correct polarity is verified before connecting any battery to the battery sockets which are connected to the battery management module.

WARNING: FIRE AND/OR CIRCUIT DAMAGE WILL RESULT FROM CONNECTING A BATTERY WITH INCORRECT POLARITY.

Installing the Socket Header Strip for the Adafruit PowerBoost 1000C Module

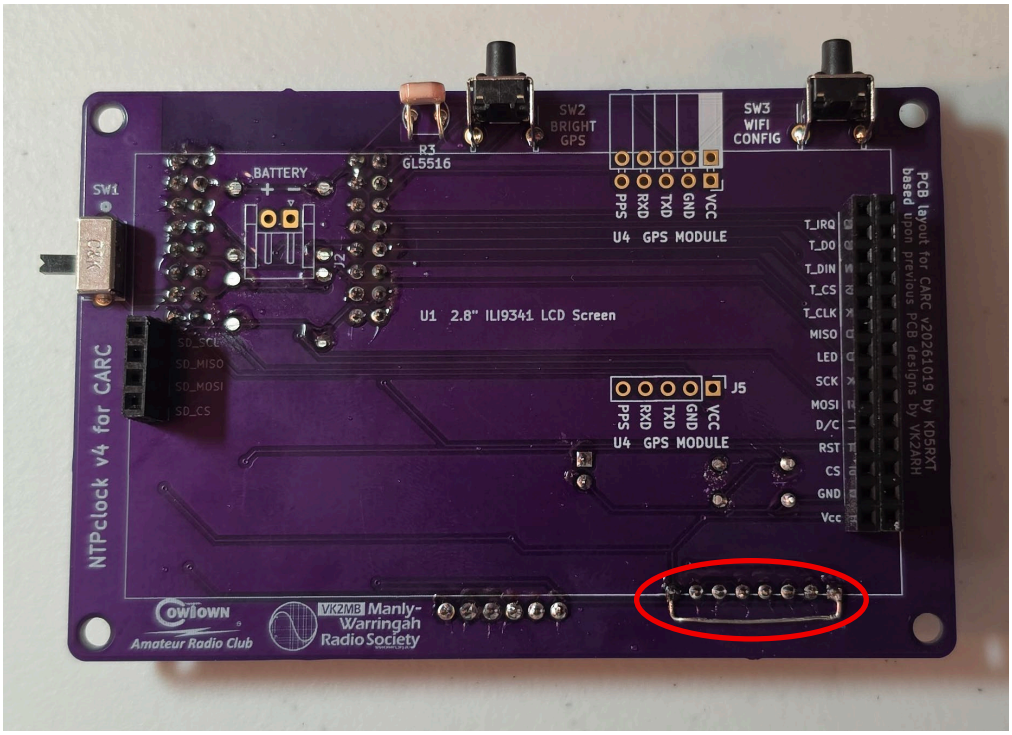


NOTE: The Adafruit PowerBoost 100C battery management module is not designed to be operated without a LiPo battery connected. Powering the Adafruit PowerBoost 1000C battery management module without a battery connected may result in damage to the module.

- [Rear of the PCB] If battery operation is desired, install a female 8-pin socket header strip for the Adafruit PowerBoost 1000C battery management module on the rear of the main PCB, as indicated by the red oval above.

Installing the LiPo Battery Sockets

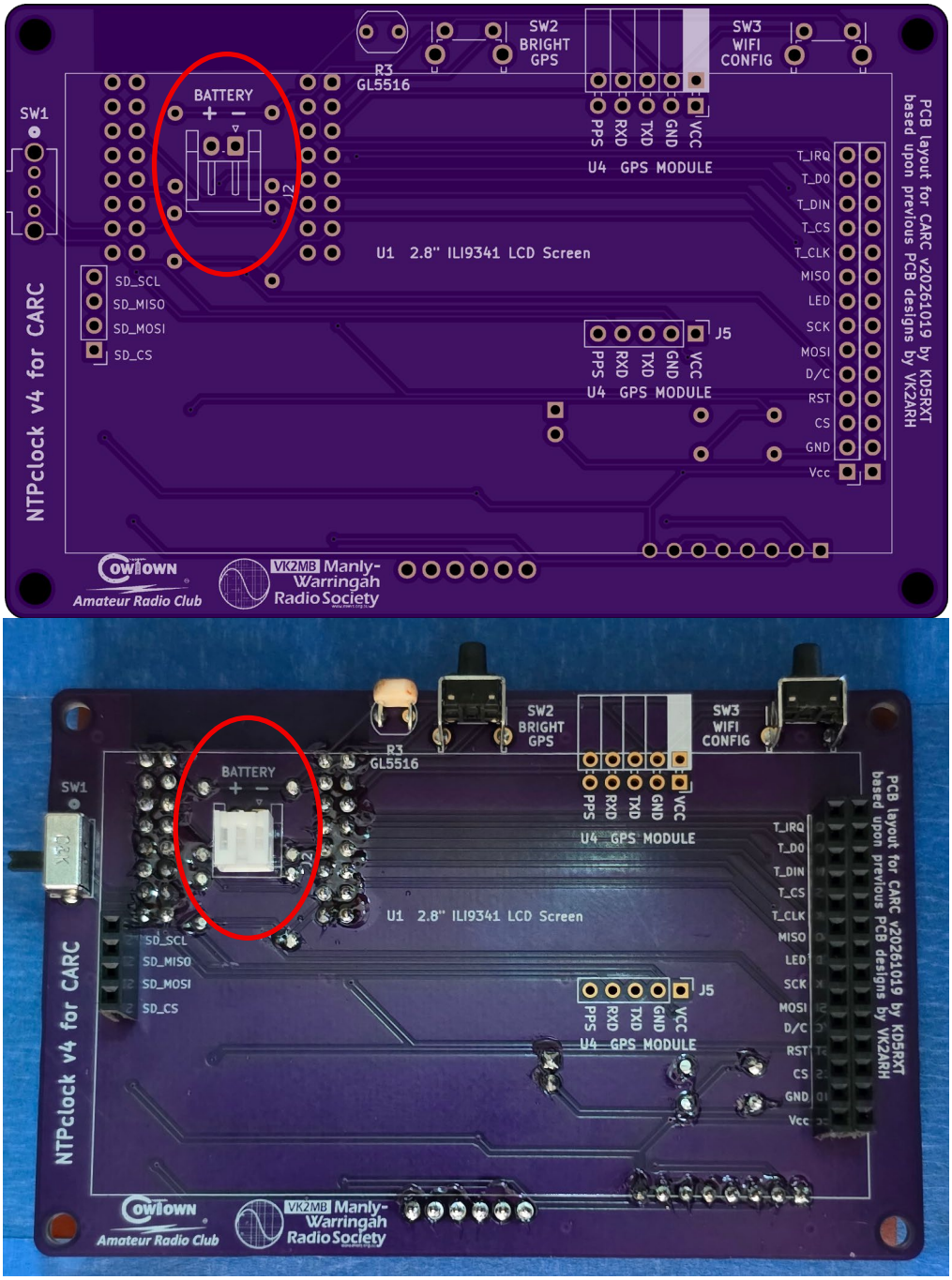
If battery operation is desired (e.g. when operating away from available USB power), then one of two installation options is available: using a 785060 (7mm thick x 50mm wide x 60mm long) 2500mAh LiPo battery from Adafruit (Product ID: 328) can be installed on the front of the PCB, with the battery's 2.0mm JST PH connector plugged into the additional battery socket, also on the front of the PCB, or using the optional 104058 (10mm thick x 40mm wide x 58mm long) 3000mAh battery (extracted from the Emergency Phone Charger from ChargeTab) can be installed on the rear of the PCB, with the battery's 2.0mm JST PH connector plugged directly into the battery socket on the Adafruit PowerBoost 1000C battery management module. A LiPo battery providing a minimum capacity of 1000mAh is recommended (note the limiting dimensions on each side of the PCB if/when selecting a different battery), where a 2500-3000mAh LiPo battery will provide 14-20 hours of operation, and should work well for most end use scenarios.



- If battery operation is not wanted and/or not needed, the additional 2.0mm JST PH connector and the Adafruit PowerBoost 1000C battery management module need not be installed. In this case, simply form a jumper using a piece of a leg trimmed from any resistor installed earlier, bent into a U-shape, and solder it in place to connect the "5Vo" pin location to the "USB" pin location on the front of the PCB where the Adafruit PowerBoost 1000C battery management module would otherwise be installed, as indicated by the red oval above. Be very careful not to contact and/or short any other pins on that connector.

  NOTE: The power on-off switch will be inoperative in this configuration, and the NTPclock will be powered continuously anytime that external 5VDC is supplied through the back panel mounted USB-C connector.

  NOTE: If, at some time in the future, you wish to make use of battery operation, it is imperative that this jumper MUST be removed before the Adafruit PowerBoost 1000C module is installed.

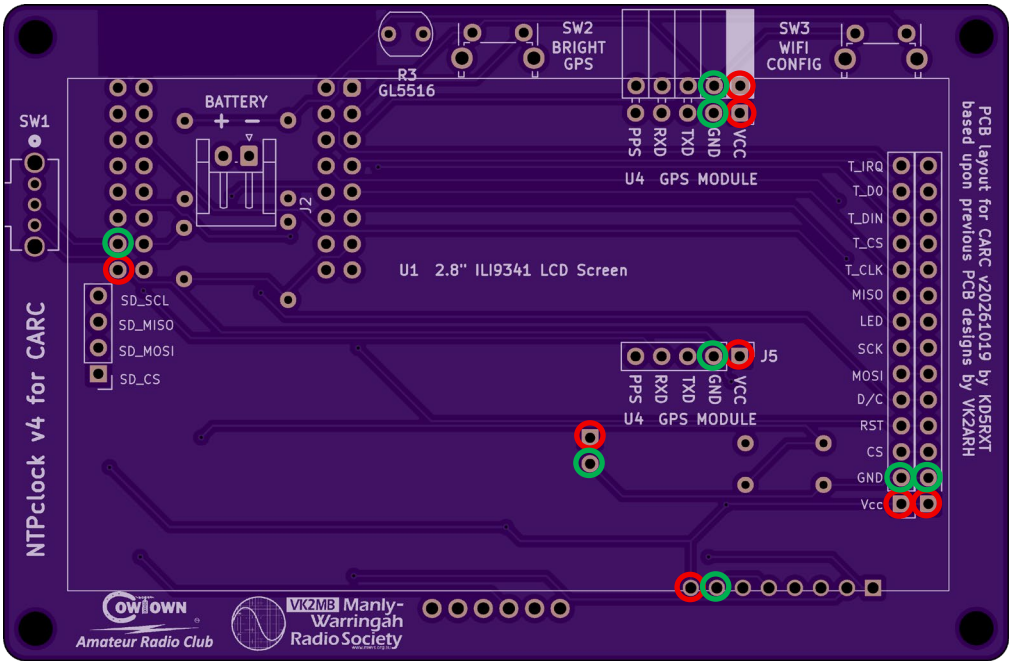


☐ [Front of the PCB] If battery operation is desired, mount the 2.0mm JST PH Connector (J2) onto the front of the PCB (solder connections are on the rear of the PCB) as indicated by the red oval above. Note the correct orientation of the socket and be sure to align the socket to correctly match the screen print on the PCB.

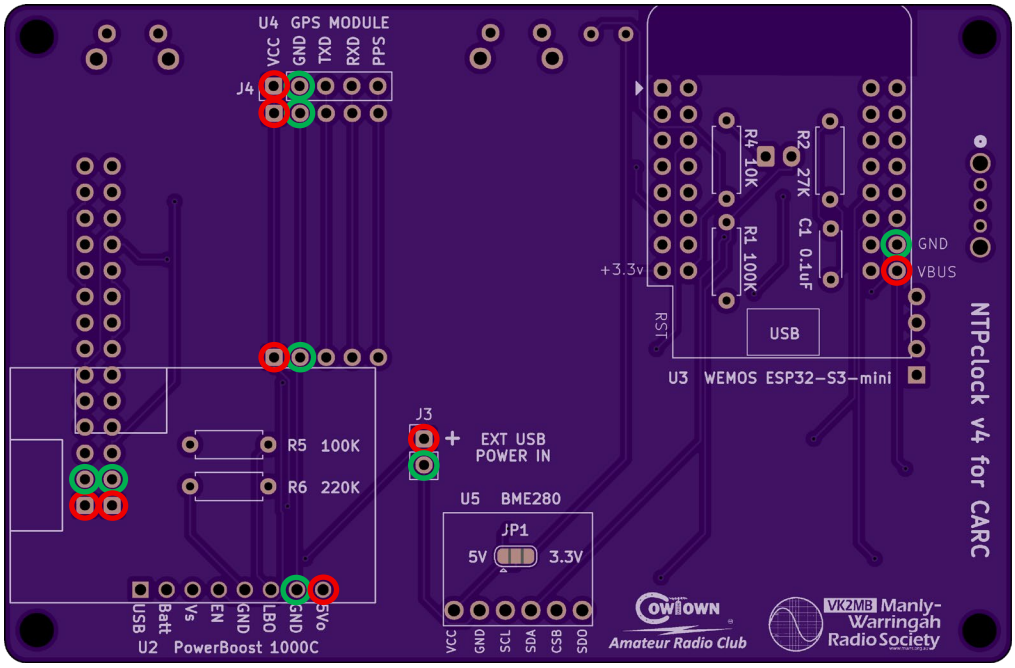
Installation is Complete !!

With the exception of installation of the GPS, as well as final assembly, this completes the build process of the NTPclock v4 for CARC (20261019) kit. At this point, you should now perform the Basic Pre-Startup Power Tests before proceeding to install the modules.

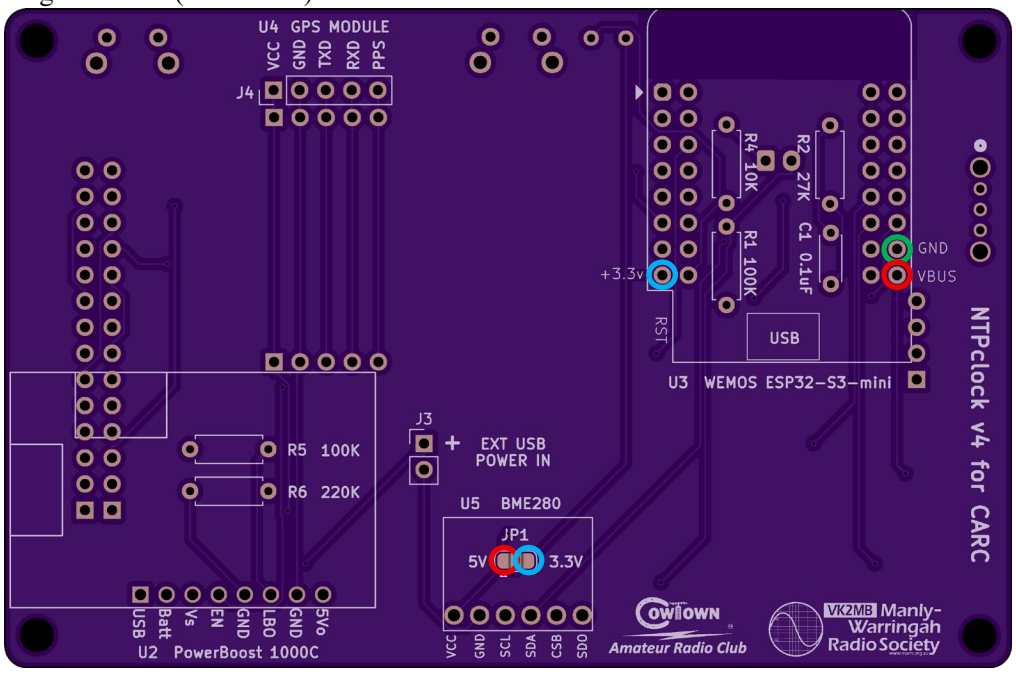
Basic Pre-Startup Power Tests



- Using a multimeter check that you have no shorts between +5VDC and Ground on the board. There are several easily accessible locations to perform this check on the front of the PCB, as indicated by the red ovals (+5VDC) and the green ovals (GROUND) above.



- There are also several easily accessible locations to perform this check on the rear of the PCB, as indicated by the red ovals (+5VDC) and the green ovals (GROUND) above.

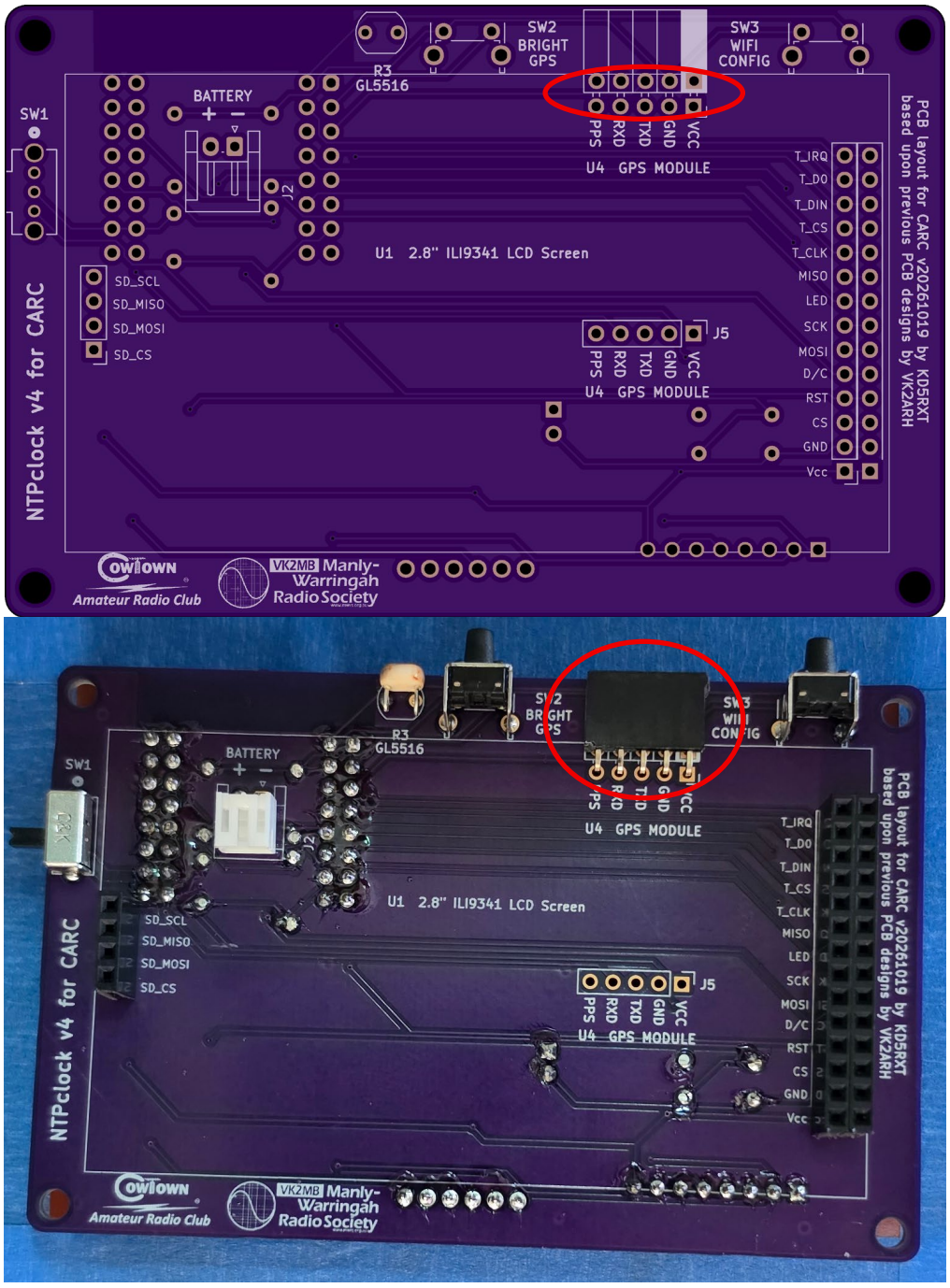


- Using a multimeter check that you have no shorts between +3.3VDC and +5VDC, and no shorts between +3.3VDC and Ground on the board. There are several easily accessible locations to perform this check on the rear of the PCB, as indicated by the aqua ovals (+3.3VDC), by the red ovals (+5VDC), and by the green ovals (GROUND) above.
- If you find a short of any kind, the cause of any short(s) must be thoroughly investigated and completely resolved before proceeding.

- Connect a USB-C 5VDC power source to the USB-C connector for the back of the enclosure and verify that 5v is being correctly supplied at the 5VDC points previously tested for shorts. Note: if support for using the Adafruit PowerBoost 1000C module was installed earlier, then the Adafruit PowerBoost 1000C battery management module will actually need to be installed & a LiPo battery properly connected (either connected directly to the Adafruit PowerBoost 1000C battery management module, or to the additional 2.0mm JST PH battery connector, as appropriate) before activating the power switch to supply 5VDC power to the board to be tested.

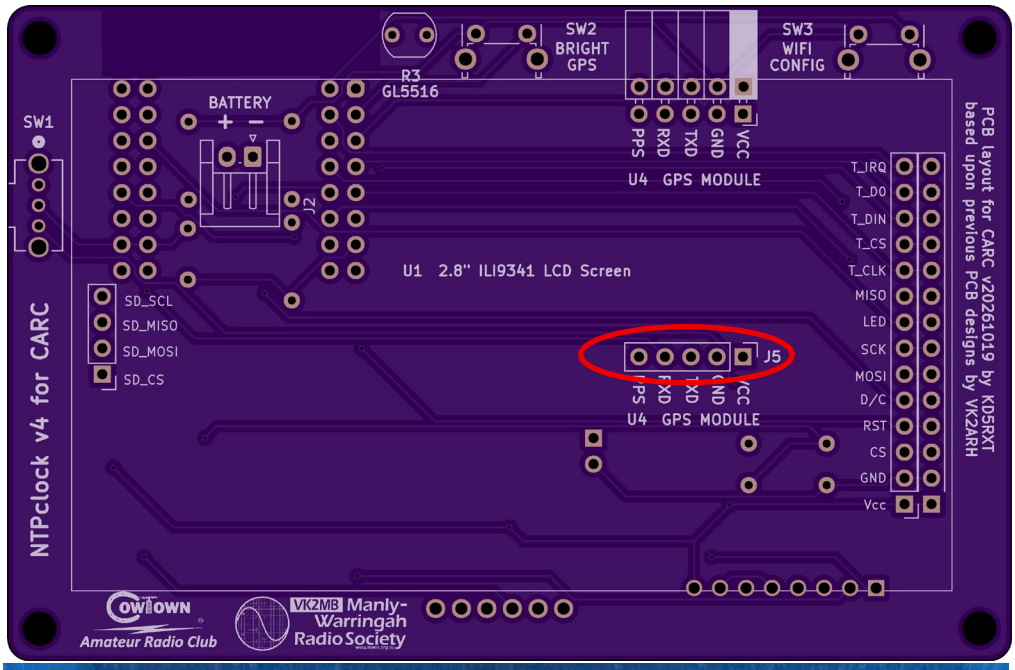
GPS Module

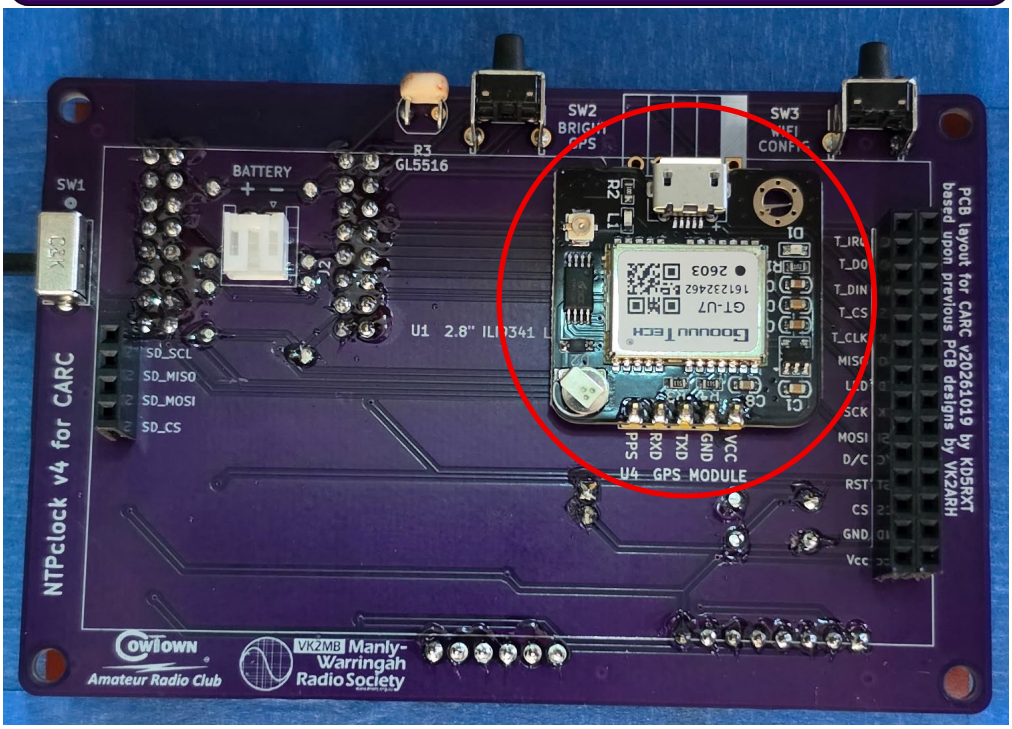
For installation of the GPS module, three options are supported by the NTPclock v4 for CARC (20261019) PCB as follows:



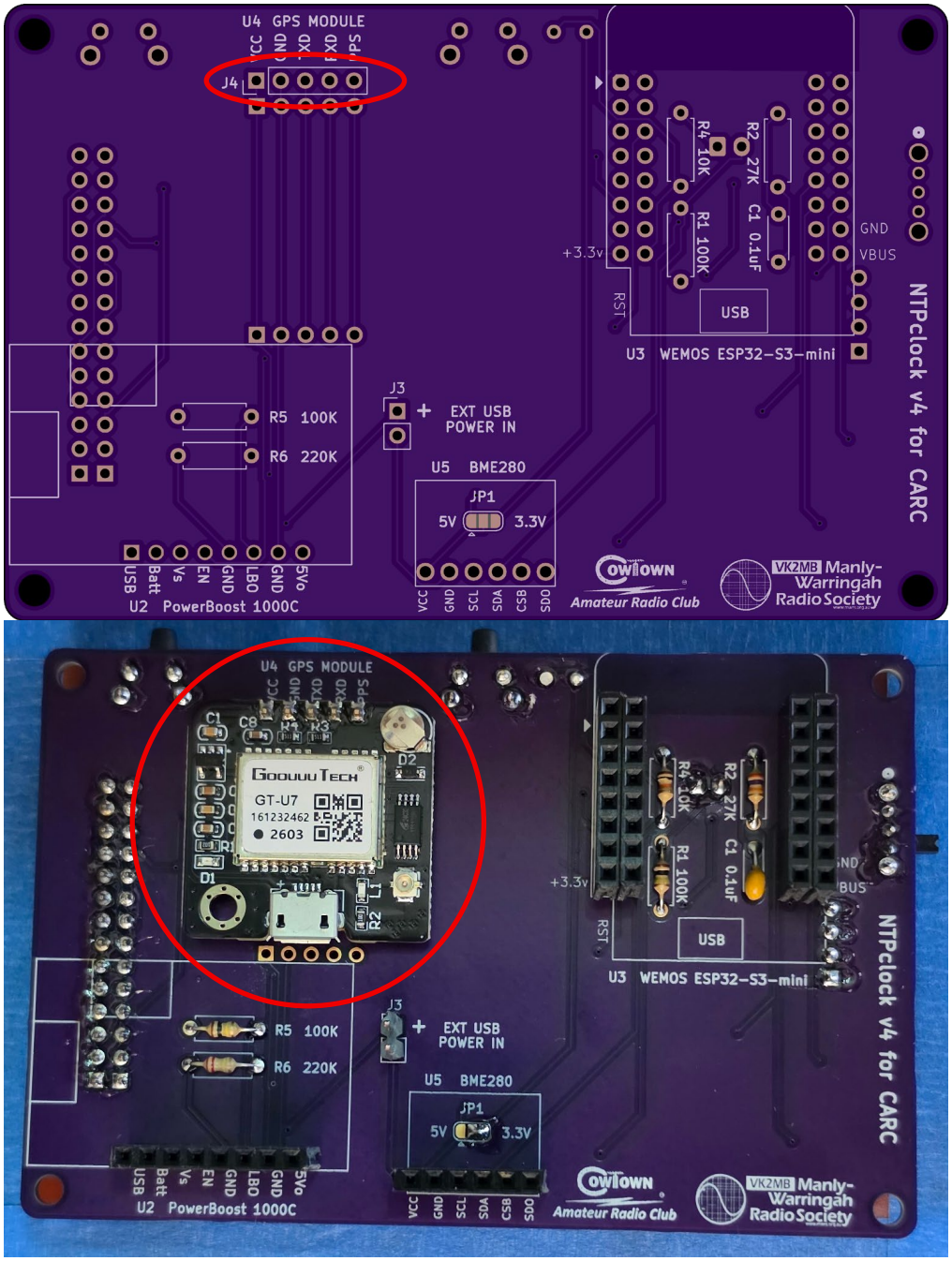
- [**OPTION #1** - Front of the PCB] For temporary installation (allowing removal of the GPS when not in use - this is the recommended installation method), the GPS module will connect to the PCB using a female 5-pin right-angle header soldered into the U4 position on the front of the PCB corresponding to one of the two female 5-pin socket header strips of the footprint silkscreen, as indicated by the red oval above. This female 5-pin socket header should be installed such that

the open end of it sits flush with the edge of the PCB. With the GPS mounted this way, the battery (either the 3000mAh LiPo battery or the 2500mAh LiPo battery, as appropriate) can be installed on either side of the PCB as described earlier.





- [**OPTION #2** - Front of the PCB] For permanent installation, where it is desirable to install & make use of the 3000mAh LiPo battery (which will be located on the rear of the PCB), the GPS module can be soldered directly into J5 (with the body of the GPS module pointing towards the top of the PCB) on the front of the PCB, as indicated by the red oval above.



☐ [**OPTION #3** - Rear of the PCB] For permanent installation, where it is desirable to install & make use of the 2500mAh LiPo battery (which will be located on the front of the PCB), the GPS module will be soldered directly into J4 (with the body of the GPS module pointing towards the bottom of the PCB) on the rear of the PCB, as indicated by the red oval above.

Regardless of which of the three options is chosen for GPS installation, in order to minimize any interference with an installed optional battery, any excess length on the legs of the GPS connector soldered into the PCB must be trimmed flush to the PCB itself using a pair of wire cutters.

GPS Antenna Considerations

Generally, GPS modules perform better when their antenna is pointing upwards, and the signal path between the satellites and the antenna is not shielded by material that might in any way attenuate the GPS signal. When operating outdoors, most GPS modules will perform well enough to support the operation of the clock, using the supplied ceramic antenna. The GT-U7 GPS module included in the kit has proved to be a solid & reliable performer, with good sensitivity and rapid satellite acquisition time, including reliable operation indoors, as long as its antenna is optimally mounted.

As indicated earlier, the best orientation for any GPS antenna is for the top of the antenna to be facing the sky. Typical GPS antennas will also work in almost any other orientation, with the exclusive exception of being mounted upside down (with the top of the antenna facing the ground). Optimal orientation should be taken into consideration as you contemplate which mounting option you may choose for the installation of the GPS module.

Final Assembly

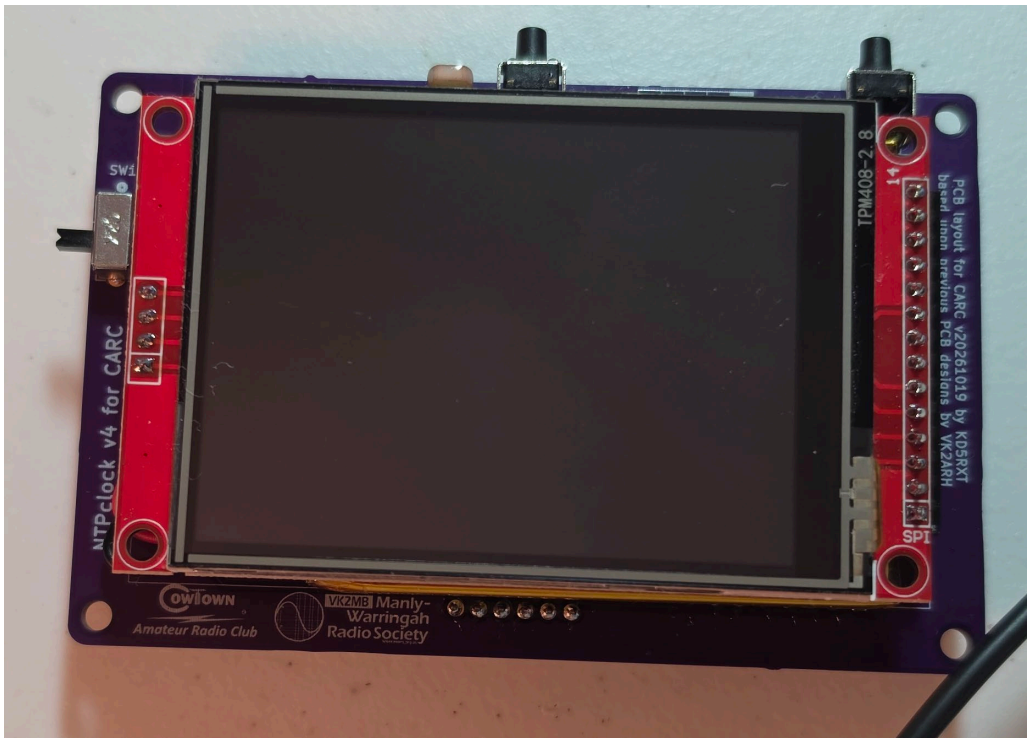
Now that all of the installation steps, as well as the basic pre-startup power tests have been completed successfully, it is now safe to install all of the NTPclock external modules (and the optional LiPo battery, if chosen for use in your kit build).



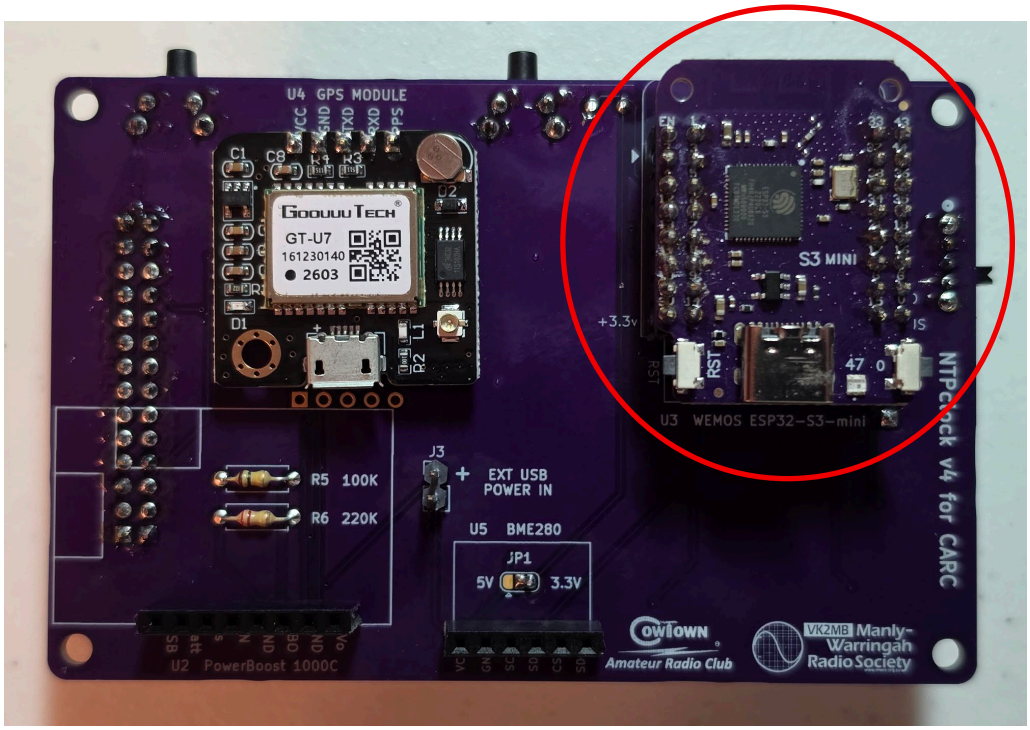
WARNING: The batteries and the battery charging module used in the NTPclock v4 for CARC (20261019) PCB follow the Adafruit convention for LiPo battery connection polarity. It is absolutely critical that correct polarity is verified before connecting any battery to the battery sockets which are connected to the battery management module.

- [ ] [Front of the PCB] For those installations where it is desirable to install & make use of the 2500mAh LiPo battery, place the battery on the front of the PCB and install the battery connector into the additional 2.0mm JST PH LiPo battery connector (also on the front of the PCB), as indicated by the red oval above.

Make sure to route the battery wires such that they neither interfere with the 4-pin display connector, nor do they run any risk of being pinched by any other components (case, standoffs, etc.) during the final assembly process.

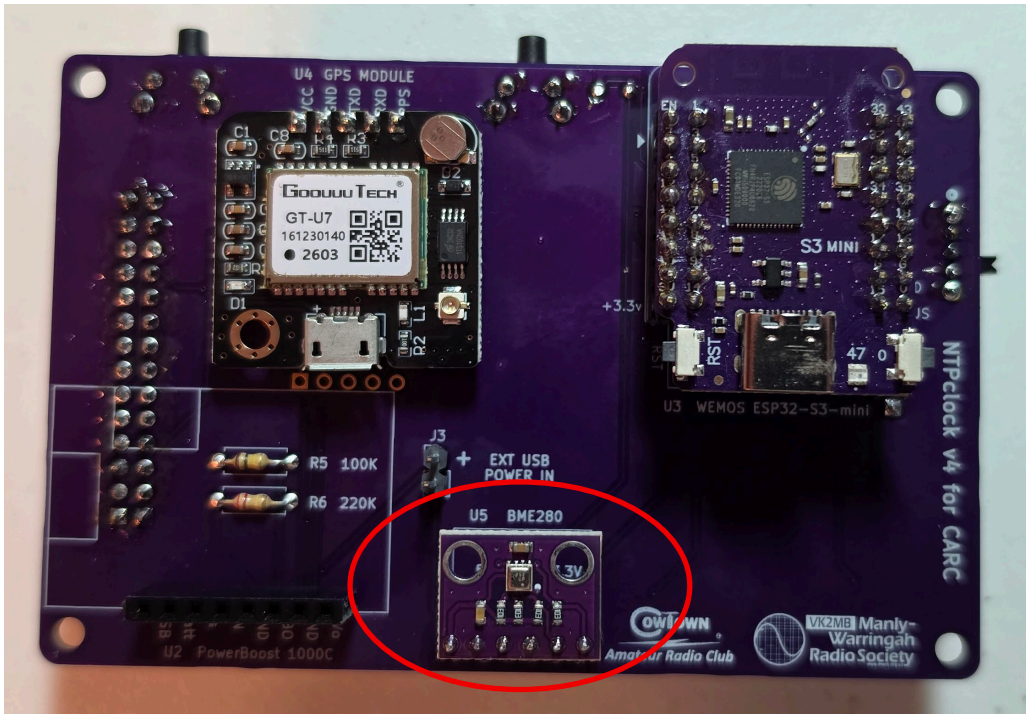


☐ [Front of the PCB] Install the ILI9341 TFT LCD display, as shown above.

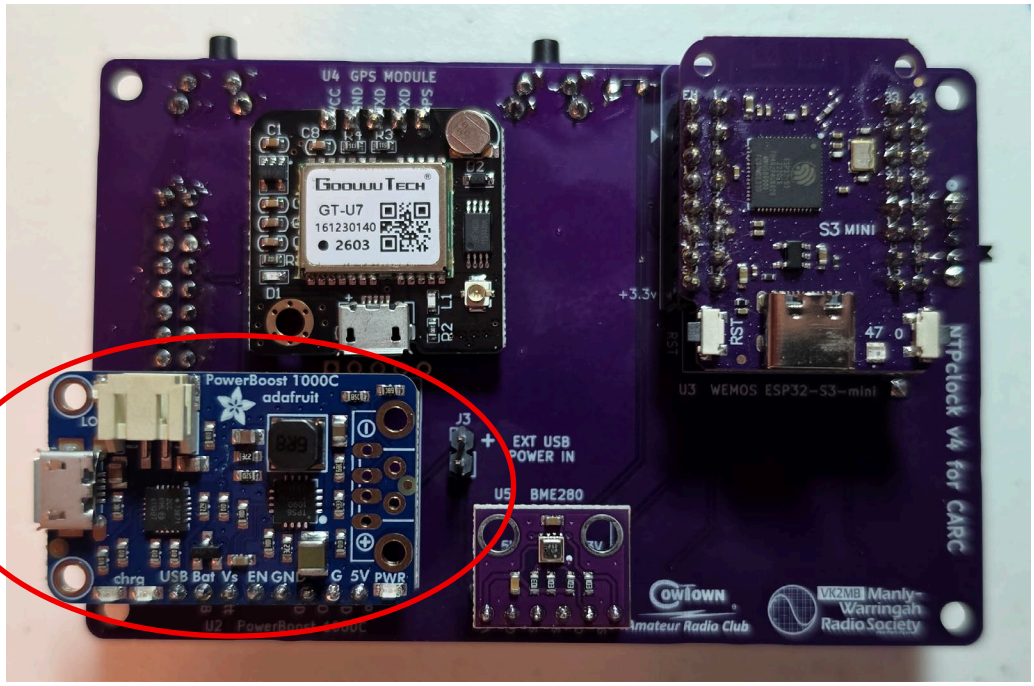


☐ [Rear of the PCB] Install the ESP32-S3-mini processor module, as indicated by the red oval above.

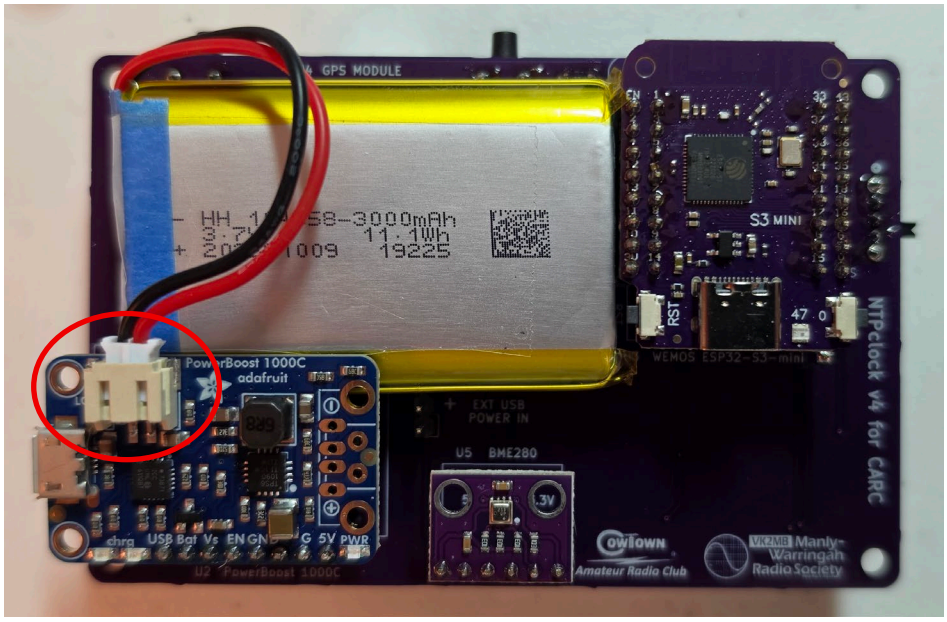
Pay particular attention to the orientation of the ESP32-S3-mini processor module.



☐ [Rear of the PCB] Install the BME280 environmental module, as indicated by the red oval above.



- [Rear of the PCB] For those installations where it is desirable to install & make use of either the 2500mAh LiPo battery (located on the front of the PCB), or the 3000mAh LiPo battery (to be located on the rear of the PCB), install the Adafruit PowerBoost 1000C battery management module, as indicated by the red oval above.

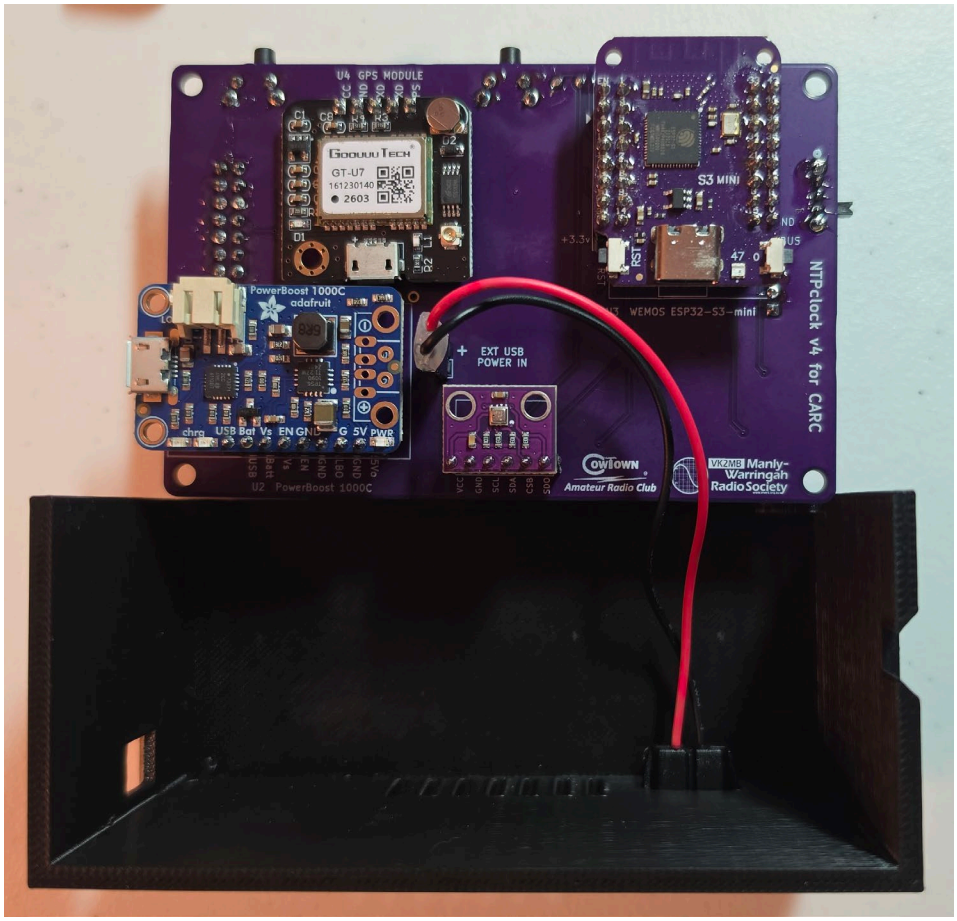


WARNING: The batteries and the battery charging module used in the NTPclock v4 for CARC (20261019) PCB follow the Adafruit convention for LiPo battery connection polarity. It is absolutely critical that correct polarity is verified before connecting any battery to the battery sockets which are connected to the battery management module.

- [Rear of the PCB] For those installations where it is desirable to install & make use of the 3000mAh LiPo battery, place the battery on the rear of the PCB and install the battery connector directly into the 2.0mm JST PH battery connector on the Adafruit PowerBoost 1000C battery management module (also on the rear of the PCB), as indicated by the red oval above.



- [Rear of the PCB] Feed the USB-C power connection cable prepared earlier through the opening in the rear-panel of the 3D-printed case, as shown above.



- Connect to 2-pin female header to the 2-pin male header installed earlier in the EXT USB POWER IN location on the rear of the PCB, as shown above.

  Pay particular attention to match the orientation between the 2-pin female header strip attached to the rear-panel USB-C connector and the orientation of the mating 2-pin male header strip soldered onto the PCB (the red wire MUST be connected ONLY to the pin marked with a "+" symbol. When connected correctly, the red wire from the rear-panel USB-C assembly will face towards the top of the PCB, and the black wire will face towards the bottom of the PCB. Damage and/or fire may occur if this connection is made incorrectly).

Final Assembly is Complete !!

Congratulations !! You have successfully completed the installation & assembly of your NTPclock kit. Enjoy !!

Software Installation on the ESP32-S3-mini processor

The ESP32-S3-mini processor must be programmed with appropriate firmware before use in the NTPclock. This programming can be done either before or after the ESP32-S3-mini is installed on the PCB. Instructions for installing software into the ESP32-S3-mini module are contained in the NTPclock software manual, which can be found in the documentation folder at:

https://github.com/mjculross/NTPclock-for-MWRS-CARC-REAST

Housing the CARC NTPclock

The primary intent with the NTPclock v4 for CARC (20261019) is to house the completed kit in a 3D-printed enclosure. If full enclosure is not desired, the clock can be assembled into a 'PCB Stack Sandwich' (an optional 3D-printed back cover), resulting in a simple, practical, and attractive finished product.

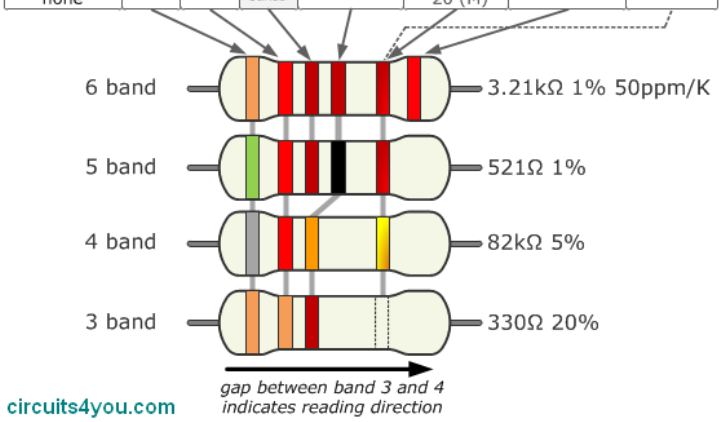
In either case, the stack sandwich is constructed in order using the following components:

- Front cover PCB (installed in front of the main PCB)
- 4 x M3 17mm F-F nylon insulated standoffs (on the front side of the main PCB)
- Main PCB
- 4 x M3 20mm M-F nylon insulated standoffs (insert the screw end of the 20mm M-F standoff through the back side of the main PCB, using the four mounting holes, and into the mating 17mm F-F standoffs - secure the two standoffs together by finger tightening only)
- Upper and lower portions of the 3D-printed enclosure are used to surround the resulting stack
- (If not employing the 3D-printed enclosure) the optional 3D-printed back cover
- 8 x M3 8mm nylon screws are screwed into the standoffs from the front and back of the 3D-printed enclosure to secure everything together

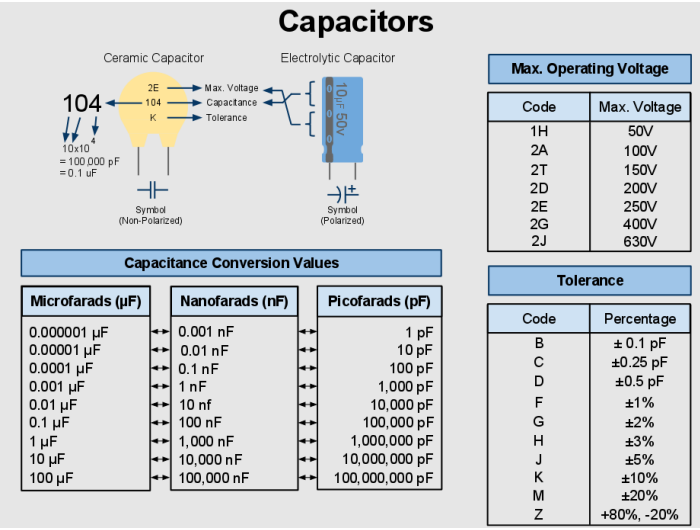
Resistor and Capacitor Identification Charts (For Info)

Resistor Color Code Chart

Color	Signficant figures			Multiply	Tolerance (%)	Temp. Coeff. (ppm/K)	Fail Rate (%)
black	0	0	0	x 1		250 (U)	
brown	1	1	1	x 10	1 (F)	100 (S)	1
red	2	2	2	x 100	2 (G)	50 (R)	0.1
orange	3	3	3	x 1K		15 (P)	0.01
yellow	4	4	4	x 10K		25 (Q)	0.001
green	5	5	5	x 100K	0.5 (D)	20 (Z)	
blue	6	6	6	x 1M	0.25 (C)	10 (Z)	
violet	7	7	7	x 10M	0.1 (B)	5 (M)	
grey	8	8	8	x 100M	0.05 (A)	1(K)	
white	9	9	9	x 1G			
gold			3th digit only for 5 and 6 bands	x 0.1	5 (J)		
silver				x 0.01	10 (K)		
none					20 (M)		



Capacitor Identification Charts and Letter Code Table



Capacitance Conversion Values

Microfarads (µF)	Nanofarads (nF)	Picofarads (pF)
0.000001 µF	0.001 nF	1 pF
0.00001 µF	0.01 nF	10 pF
0.0001 µF	0.1 nF	100 pF
0.001 µF	1 nF	1,000 pF
0.01 µF	10 nf	10,000 pF
0.1 µF	100 nF	100,000 pF
1 µF	1,000 nF	1,000,000 pF
10 µF	10,000 nF	10,000,000 pF
100 µF	100,000 nF	100,000,000 pF

Max. Operating Voltage

Code	Max. Voltage
1H	50V
2A	100V
2T	150V
2D	200V
2E	250V
2G	400V
2J	630V

Tolerance

Code	Percentage
B	± 0.1 pF
C	±0.25 pF
D	±0.5 pF
F	±1%
G	±2%
H	±3%
J	±5%
K	±10%
M	±20%
Z	+80%, -20%

Capacitor Letter Codes Table

Picofarad (pF)	Nanofarad (nF)	Microfarad (uF)	Code	Picofarad (pF)	Nanofarad (nF)	Microfarad (uF)	Code
10	0.01	0.00001	100	4700	4.7	0.0047	472
15	0.015	0.000015	150	5000	5.0	0.005	502
22	0.022	0.000022	220	5600	5.6	0.0056	562
33	0.033	0.000033	330	6800	6.8	0.0068	682
47	0.047	0.000047	470	10000	10	0.01	103
100	0.1	0.0001	101	15000	15	0.015	153
120	0.12	0.00012	121	22000	22	0.022	223
130	0.13	0.00013	131	33000	33	0.033	333
150	0.15	0.00015	151	47000	47	0.047	473
180	0.18	0.00018	181	68000	68	0.068	683
220	0.22	0.00022	221	100000	100	0.1	104
330	0.33	0.00033	331	150000	150	0.15	154
470	0.47	0.00047	471	200000	200	0.2	254
560	0.56	0.00056	561	220000	220	0.22	224
680	0.68	0.00068	681	330000	330	0.33	334
750	0.75	0.00075	751	470000	470	0.47	474
820	0.82	0.00082	821	680000	680	0.68	684
1000	1.0	0.001	102	1000000	1000	1.0	105
1500	1.5	0.0015	152	1500000	1500	1.5	155
2000	2.0	0.002	202	2000000	2000	2.0	205
2200	2.2	0.0022	222	2200000	2200	2.2	225
3300	3.3	0.0033	332	3300000	3300	3.3	335